Egocentric Audio-Visual Understanding:
From Multimodal Signal Processing to Wearable Spatial Intelligence

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Outline

- Background
- Ego-centric long-form audio activity understanding
  - MultiAct datasets (v1, v2)
  - Hierarchical SlowFast modelling framework
  - Hierarchical grammar-tree parsing
- 360° video to spatial audio generation
  - YT-Speech dataset & a localizer-renderer model
  - AV-QTSS-Syn dataset & AV-QTSS model
- Ego-centric audio-visual understanding
  - A new dataset & benchmark
- Conclusion and future works

Egocentric Audio-Visual Understanding (EAVU)

- Concerning the **perception** and **understanding** of the surrounding world from the **first-person perspective** of a human wearer
- Using **wearable devices** such as smart glasses, head-mounted cameras, earbuds and other body-worn sensors
- Focusing on **dynamic**, **personal**, and **interactive** relationship between the wearer and their environment, different from conventional AV understanding which often assumes **fixed** cameras and microphones observing an external scene.
- Aiming to understand:
  - **what the wears sees** (e.g., objects, people, activities, visual scenes and events), **hears** (e.g., speech, music, environmental sounds, and sound events), **needs** (e.g., context-aware assistance, interaction and decision support), and **is doing** (e.g., actions, interactions, intentions and activities);
  - **where things are** (e.g., the spatial locations, directions, distances and movements of sound sources and visible objects);
  - **what is happening** (e.g., the relationship between people, objects, events, and the environment).

Typical Applications

- Assistive intelligence and accessibility
- Personal assistants
- Spatial navigation and environmental awareness
- Industrial assistance and human-machine interaction
- Sports, training and skill development
- Social interaction and communication

Main Research Areas

From **sensing and recognising** with heterogeneous modalities -> to jointly **understanding** the physical environment -> to **reasoning and interacting** intelligently within that environment:

Multimodal sensing, perception, and signal processing: Extracting reliable information from wearable sensors.

- Robust wearable audio enhancement, source separation, and event detection.
- Robust wearable visual perception, object detection, activity recognition, gaze and head pose estimation.
- Audio-visual fusion, mapping, and representation learning.
- Multimodal synchronisation, alignment, and sensor calibration.

Audio-visual scene and spatial understanding: Understanding what is happening and where it is happening.

- Sound-source localisation and tracking, audio-visual object grounding.
- 3D scene reconstruction and understanding, interaction-aware scene understanding.
- Spatial-temporal event understanding, intent recognition, and interaction-aware scene understanding
- Audio-visual action understanding, sound-producing action recognition

Wearable spatial intelligence: Building an intelligent, interactive and context-aware wearable agent.

- Multimodal foundation models, e.g., audio-language, vision-language, and audio-vision-language models.
- Spatial-temporal reasoning, grounded audio-visual reasoning, and personalised reasoning.
- Long-form multimodal understanding, reasoning, and question answering.
- Personalised perception, active perception and interaction, context-aware assistance.
- Human-AI interaction/collaboration, and people centred AI agents.
- Efficient modelling, on-device inference, adaptive sensing and power management.

Open Challenges

Challenges	Key research questions
1. Multimodal alignment	How can asynchronous audio, video, gaze and IMU signals be accurately aligned?
2. Cross-modal grounding	How can sounds be reliably associated with their visible sources?
3. Dynamic spatial perception	How can spatial information remain consistent during head and body movement?
4. Ego-motion compensation	How can the system distinguish movement of the wearer from movement of external sources?
5. Robust perception	How can the system operate under noise, occlusion, reverberation and adverse weather?
6. Long-form understanding	How can important events be retained, indexed and retrieved over extended periods?
7. Generalisation	How can models generalise across people, devices, environments and activities?
8. Real-time processing	How can sophisticated multimodal reasoning operate within wearable computational and energy constraints?
9. Privacy and trustworthiness	How can continuous sensing protect personal information and avoid inappropriate inference?
10. Evaluation	How can we evaluate spatial understanding, reasoning, interaction and real-world usefulness?

Datasets

- **Charades-Ego** (2018, Paired first- and third-person videos; 34.4 hours)
- **Ego-CH** (2020, RGB; 27 hours)
- **EgoCom** (2020, audio, video, text; 38.5 hours)
- **EPIC-KITCHEN/EPIC-SOUND** (2020/2023, RGB, audio, text; 100 hours)
- **Ego4D** (2021, RGB, partial audio, gaze, 3D; >3600 hours)
- **EasyCom** (2021, audio, video, head-pose, speech transcription; 5 hours)
- **MECCANO** (2021, RGB, hand-object annotations; 6.9 hours)
- **Assembly101** (2022, RGB, 3D annotations; 513 hours)
- **Ego-Exo4D** (2023, audio, video, IMU, gaze, 3D; 1286 hours)
- **Ego-AVSL*** (2023, audio, video; 10 hours)
- **WEAR** (2023, RGB, IMU; 15 hours)
- **HoloAssist** (2023, RGB, depth, eye gaze, hand/head pose, audio; 169 hours)
- **EGOFALLS** (2023, audio, video; 6 hours)
- **Ego4DSounds** (2024, audio, video, texts; >950 hours)
- **HD-EPIC** (2025, RGB, audio, gaze, SLAM; 41.3 hours)

* **Ego-AVSL** is a dataset that we created using Unity game engine augmented by ResonanceAudio for AR/VR applications, with speech samples taken from LibriSpeech datasets. Details can be found from:
J. Zhao, Y. Xu, X. Qian, W. Wang, “Audio visual speaker localization from egocentric views”, 2023, https://arxiv.org/abs/2309.16308.
Data & code: https://github.com/KawhiZhao/Egocentric-Audio-Visual-Speaker-Localization

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◆ Research Motivation

The Critical Gap: From Short Events to Long-form Activities in Audio-only Domain:

Dataset	Scale (#hours)	Avg. Duration (s)	Event-level	Sub-activity	Activity	Hierarchical Structure
VGGSound	~550	10s (fixed)	✓	✗	✗	✗
Clotho	~24	15~30s	(Captions)	✗	✗	✗
AudioSet	~5,800	10s (fixed)	✓	✗	✗	✗
EPIC-SOUNDS	~100	~4.9s	✓	✗	✗	✗

The Problem: Focus on Isolated, Short-term Events.

- As shown above, existing audio datasets primarily consist of short, isolated sound events (e.g., 10-second clips of a 'dog bark' or 'car horn').
- This leaves a critical gap: there are no benchmarks for understanding **long-form audio activities** that are composed of a structured sequence of steps.

◆ Research Motivation

Inspired by the Ecological Psychology Framework, human activities can be understood at three levels:

Behavior Setting ← Behavior Episodes ← Atomic Physical Units

Long-from Audio-Visual Stream:



Activity ← ***Sub-activity*** ← ***Event***

Understand Hierarchy



Kitchen Scenario

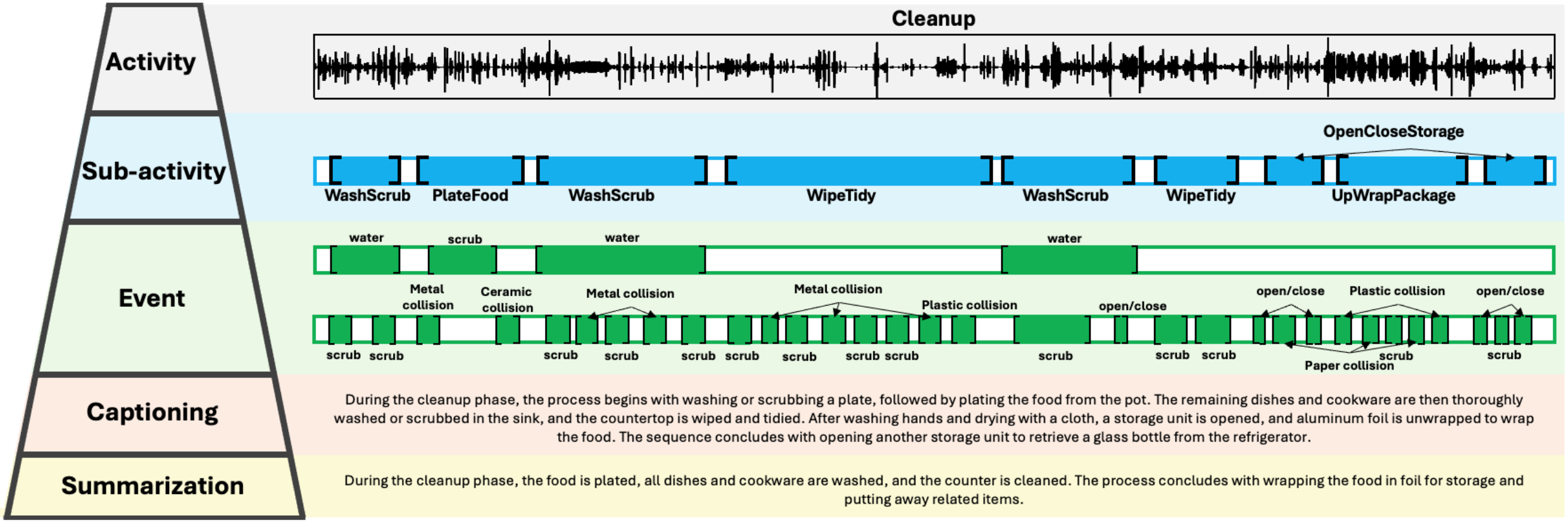
Model Audio-Visual Composition

Fine-grained Captioning

High-level Summarization

Infer User-Aware Meaning

◆ *MultiAct* Dataset



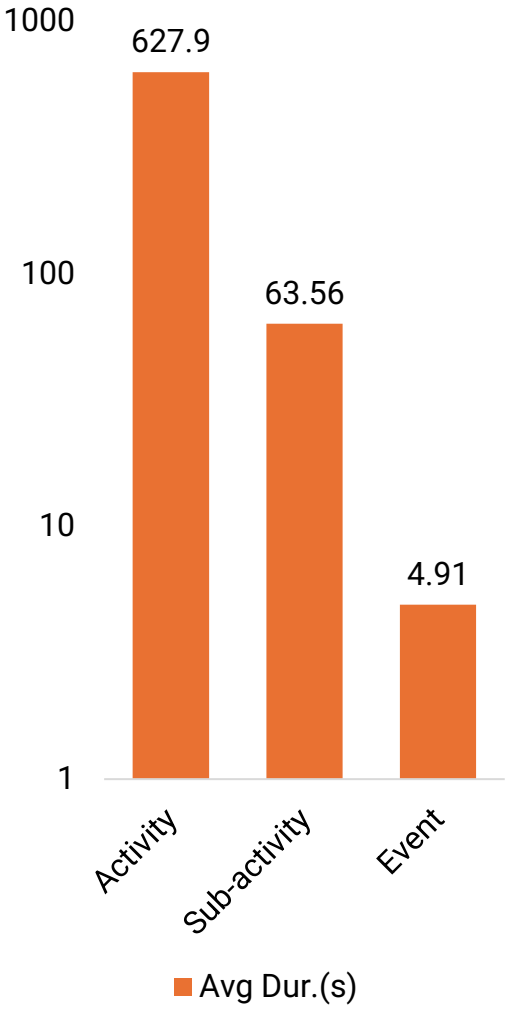
MultiAct provides dense, multi-layered annotations for kitchen activities from EPIC-SOUNDS Dataset.

P. Zhang, Q. Luo, P. Jackson, and W. Wang, "Hierarchical Activity Recognition and Captioning from Long-form Audio," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing* (ICASSP 2026), Barcelona, Spain, May 4-8, 2026. [PDF]

◆ Dataset Statistics

Key Metrics

- **Source:** 17 kitchens from EPIC-SOUNDS
- **Scale:** 8.97 hours of audio
- **Density:** Over 7,800 hierarchical annotations
- **Split:** Train (6.24h) / Val (1.14h) / Eval (1.60h)

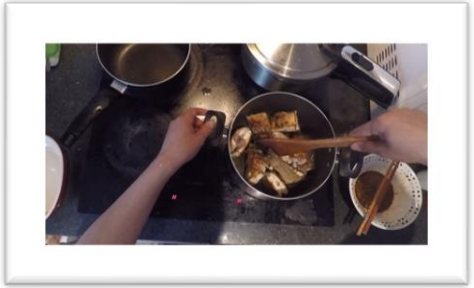


Hierarchical Details: From Minutes to Seconds

Level	#Classes	#Annotations	Duration Range	Avg Duration
Activity	3	51	15.72s ~ 3164.33s	627.90s
Sub-activity	12	472	0.57s ~571.96s	63.56s
Event	44	7312	0.20s ~ 100.02s	4.91s

Evolution of the MultiAct Dataset

MultiAct v1.0



- **Source:** 17 kitchens from EPIC-KITCHENS

Scene	Total Dur. (h)	Dur. Range (min)
Kitchen-only	8.97	8.7 ~ 61.8

MultiAct v2.0

- **Source:** curated from Ego4D and EPIC-KITCHENS

Scene	Total Dur. (h)	Dur. Range (min)
Carpentry	25.96	3.0 ~ 85.8
Cooking	37.18	3.1 ~ 62.2
Household	37.74	3.0 ~ 76.8
Overall	100.88	3.0 ~ 85.8



Carpentry



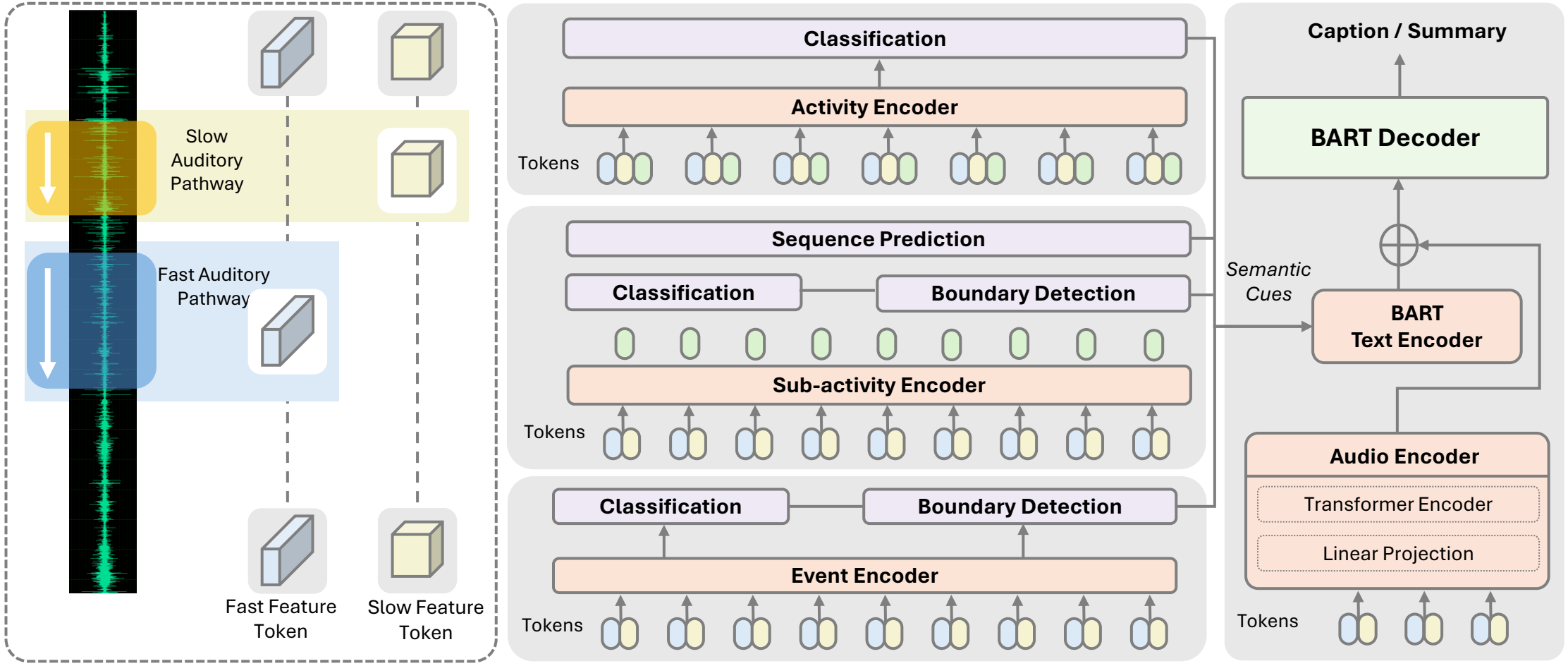
Household



Cooking

◆ A Hierarchical SlowFast Model

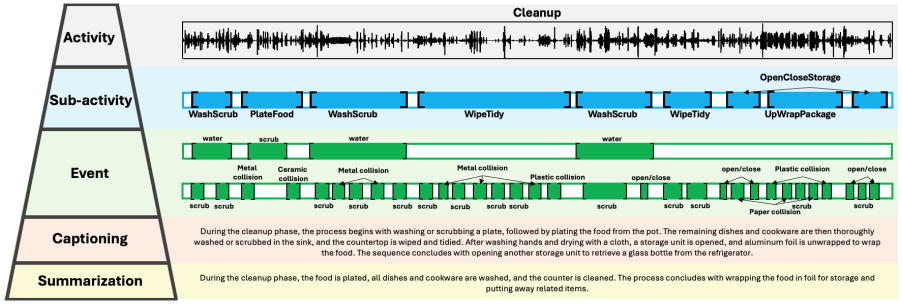
A Hierarchical SlowFast Model for Long-form Audio Activity Recognition and Captioning



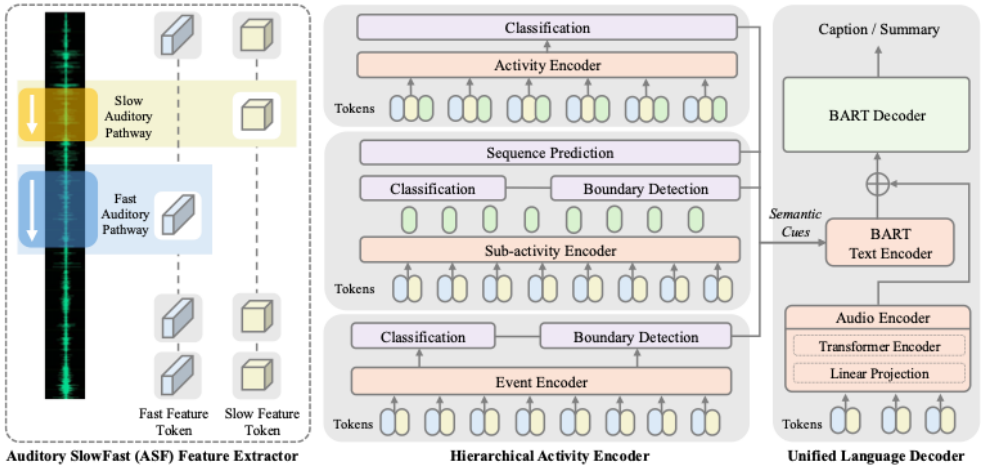
Auditory SlowFast (ASF) Feature Extractor **Hierarchical Activity Encoder** **Unified Language Decoder**

◆ Experiment

Our Contributions



MultiAct Dataset



Hierarchical Modeling Method

Benchmark Tasks

Audio Activity Recognition

- Audio Activity Classification
- Audio Activity Detection
- Audio Activity Sequence Prediction

Audio Activity Captioning

◆ Audio Activity Classification

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- **Result & Analysis**

Split	Level	Model	Top-1% (↑)	Top-5 % (↑)	mPCA % (↑)	mAP % (↑)	mAUC % (↑)
Val	Event	ASF	69.0	93.6	30.1	41.6	93.5
	Sub-activity	ASF-Atten	50.0	**92.6**	55.2	**64.9**	85.5
		ASF-CrossAtten	**51.9**	88.9	**49.7**	65.9	**87.5**
	Activity	ASF-Atten	60.0	N/A	60.7	**79.4**	87.2
		ASF-CrossAtten	**66.7**	N/A	**61.9**	72.7	84.6
Eval	Event	ASF	67.2	92.4	33.1	41.5	91.9
	Sub-activity	ASF-Atten	**51.3**	**83.3**	**47.4**	**41.9**	**77.5**
		ASF-CrossAtten	42.3	74.4	33.8	37.0	76.4
	Activity	ASF-Atten	83.3	N/A	83.3	**94.4**	**95.8**
		ASF-CrossAtten	**83.3**	N/A	**83.3**	72.2	70.8

** Top-5 is not applicable (N/A) for the Activity level as there are fewer than 5 classes.*

- **MultiAct Presents Significant Challenges:** The absolute performance, especially for fine-grained Sub-activity and Event classification (e.g., mAP < 70%), indicates that MultiAct is a highly challenging benchmark.

- **Opportunities for Future Research:** There is significant room for improvement in modeling temporal context and hierarchical dependencies.

◆ Audio Activity Detection

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- **Result & Analysis**

Split	Level	Average Precision (AP) at different IoU Thresholds (%, ↑)					
		@0.1	@0.2	@0.3	@0.4	@0.5	Mean
Val	Event	17.0	14.7	12.8	11.2	9.8	13.1
	Sub-activity	44.3	41.0	30.8	25.3	24.3	33.1
Eval	Event	16.5	15.7	15.0	13.6	12.5	14.6
	Sub-activity	41.8	37.3	32.2	26.0	22.0	31.9

- **The challenge of precise localization:** The sharp performance drop as the IoU threshold increases (from 41.8% @0.1 to 22.0% @0.5) reveals that while the model can roughly locate events, achieving precise boundary detection is extremely difficult.
- **Dense, overlapping events & subtle cues:** The primary challenge stems from the dataset's nature, which features dense clusters of overlapping events and subtle acoustic cues that are hard to distinguish.
- **Varying temporal scales of sub-activities:** The difficulty is further compounded by the wide variation in duration across different sub-activities.

◆ Audio Activity Sequence Prediction

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- **Result & Analysis**

Split	Decode Chunk	AER% on Training Context Length (Number of Activities) (%, ↓)					
		2	**3**	**4**	**6**	**8**	**Full**
Val	Fix Chunk	**59.0**	64.1	74.2	78.4	81.5	83.3
	Sliding Window	**66.7**	72.2	75.9	81.5	88.9	79.6
Eval	Fix Chunk	**67.4**	70.5	74.8	78.3	82.6	88.5
	Sliding Window	**69.2**	74.4	75.6	87.2	87.2	80.8

- **Local Context is Key**: Directly training on the full, long-form audio sequence yields poor performance (AER > 80%). Performance significantly improves when the model is trained on shorter contexts (e.g., chunks of 2-3 sub-activities), demonstrating that focusing on local procedural flow is a more effective strategy for this task.

- **A Challenging Frontier**: Even with the optimized training strategy, the best achieved Activity Error Rate remains high (AER ~67%). This highlights that audio activity sequence prediction is a highly challenging frontier. Accurately modeling the complex, long-range dependencies and variations in real-world procedural audio is a significant open problem.

◆ Audio Activity Captioning

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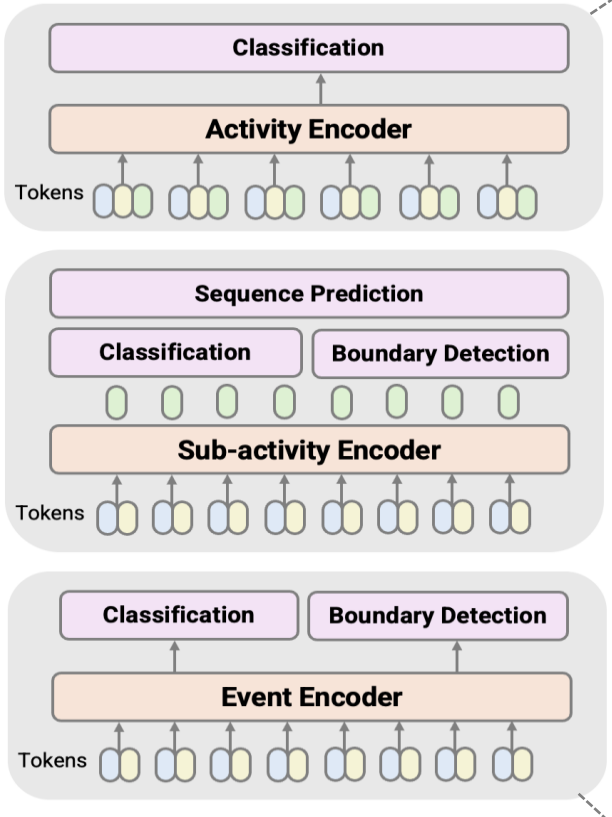
- **Result & Analysis**

Split	Task	Method	$BLEU_1$(%, ↑)	$BLEU_4$(%, ↑)	METEOR(%, ↑)	$ROUGE_L$(%, ↑)	$CIDEr$(%, ↑)
SOTA on AudioSet (Short-form Captioning, ~10s)			**70.7**	**30.9**	**25.0**	**40.0**	**78.7**
Val	*Fine-grained Captioning*	Rule	24.0	4.9	11.4	20.0	3.8
		Hierarchical	**39.2**	**10.8**	**15.8**	**28.5**	**16.2**
	High-level Summarization	Rule	23.7	8.0	11.7	26.9	9.3
		Hierarchical	**28.4**	**9.8**	**13.5**	**32.5**	**24.1**
Eval	*Fine-grained Captioning*	Rule	16.8	2.7	**13.0**	20.2	2.8
		Hierarchical	**17.8**	**3.4**	9.6	**23.0**	**20.1**
	High-level Summarization	Rule	17.3	**7.5**	8.8	20.7	2.2
		Hierarchical	**21.8**	7.3	**11.0**	**28.3**	**11.1**

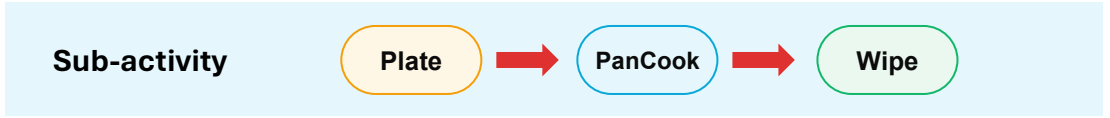
- **Hierarchical Guidance Boosts Performance**: Our Hierarchically-guided model, which leverages a dual-encoder BART architecture, consistently and substantially outperforms the Rule-based baseline across all metrics. This demonstrates the superiority of **end-to-end generative modeling over template-based methods** for complex audio captioning.

- **A Challenging Frontier**: Despite the improvements, **generating long-form, fine-grained captions for audio remains a formidable challenge.** The current performance indicates significant room for future exploration in areas such as improving factual consistency, capturing richer temporal relations, and enhancing captioning coherence.

Limitations of Layer-wise Hierarchical Inference

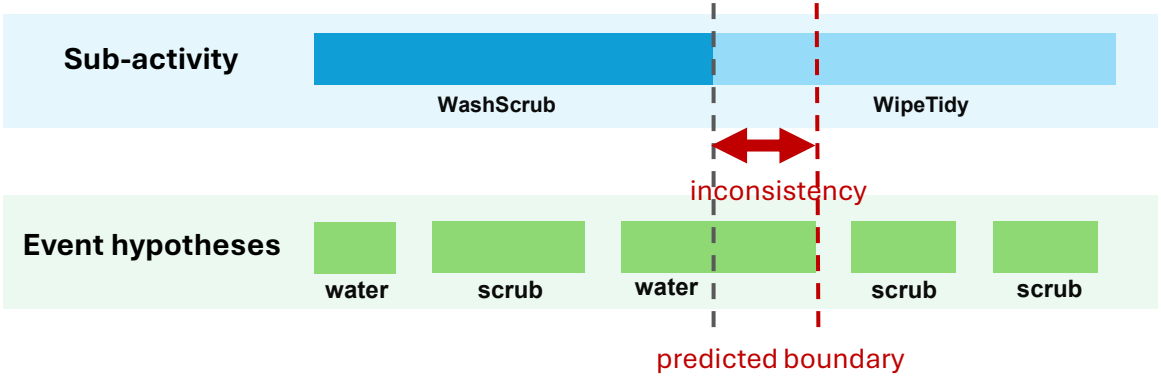


- **Order violation:** locally likely steps can break procedural order.



X

- **Temporal boundary inconsistency:** independently predicted layers may place related boundaries at different time steps. Without cross-level alignment, their segmentations can disagree and fail to form one coherent hierarchy.



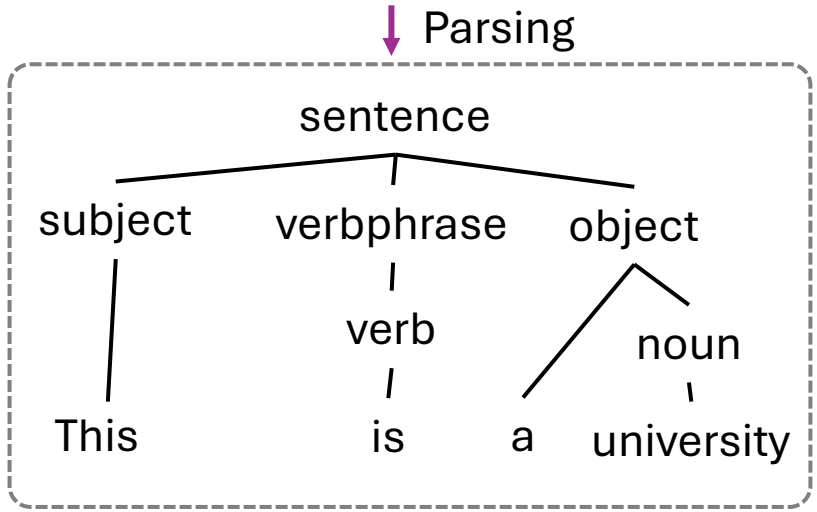
P. Zhang, Q. Luo, P. J. B. Jackson and W. Wang, " Grammar-Guided Hierarchical Parsing for Long-form Audio Activity Recognition," *Interspeech*, Sydney, Australia, 2026.

Grammar Provides the Missing Global Constraints

Production rules of an example English grammar:

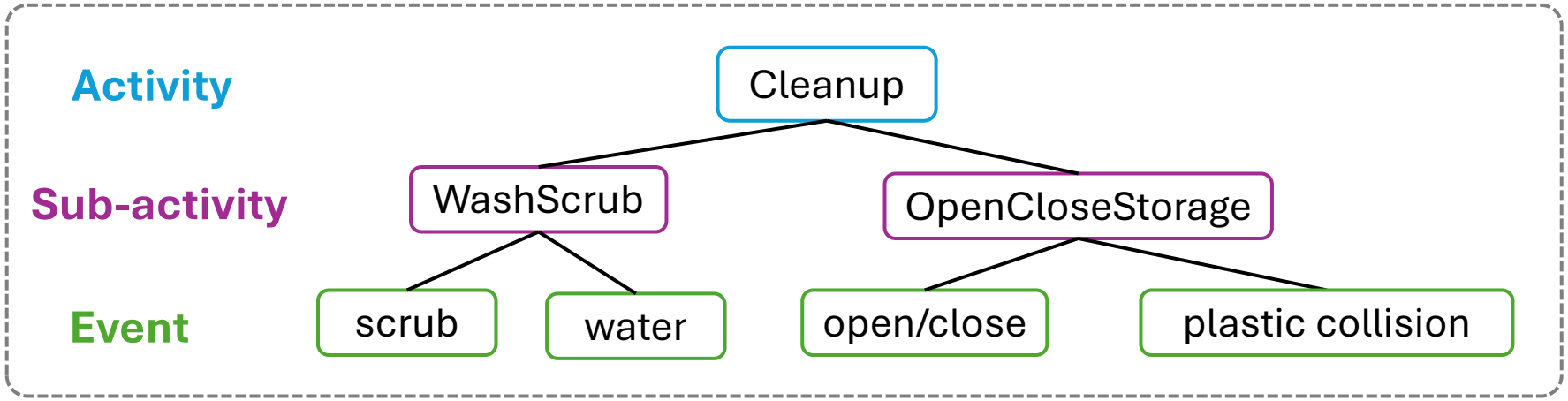
$$
\begin{aligned}
&< sentence > \quad \rightarrow \quad < subject >< verbphrase >< object > \\
&< subject > \quad \rightarrow \quad \text{“This”} \\
&< verb > \quad \rightarrow \quad \text{“is” | “run” | “am” | “tell”} \\
&< noun > \quad \rightarrow \quad \text{“university” | “world” | “lies”}
\end{aligned}
$$

“This is a university”

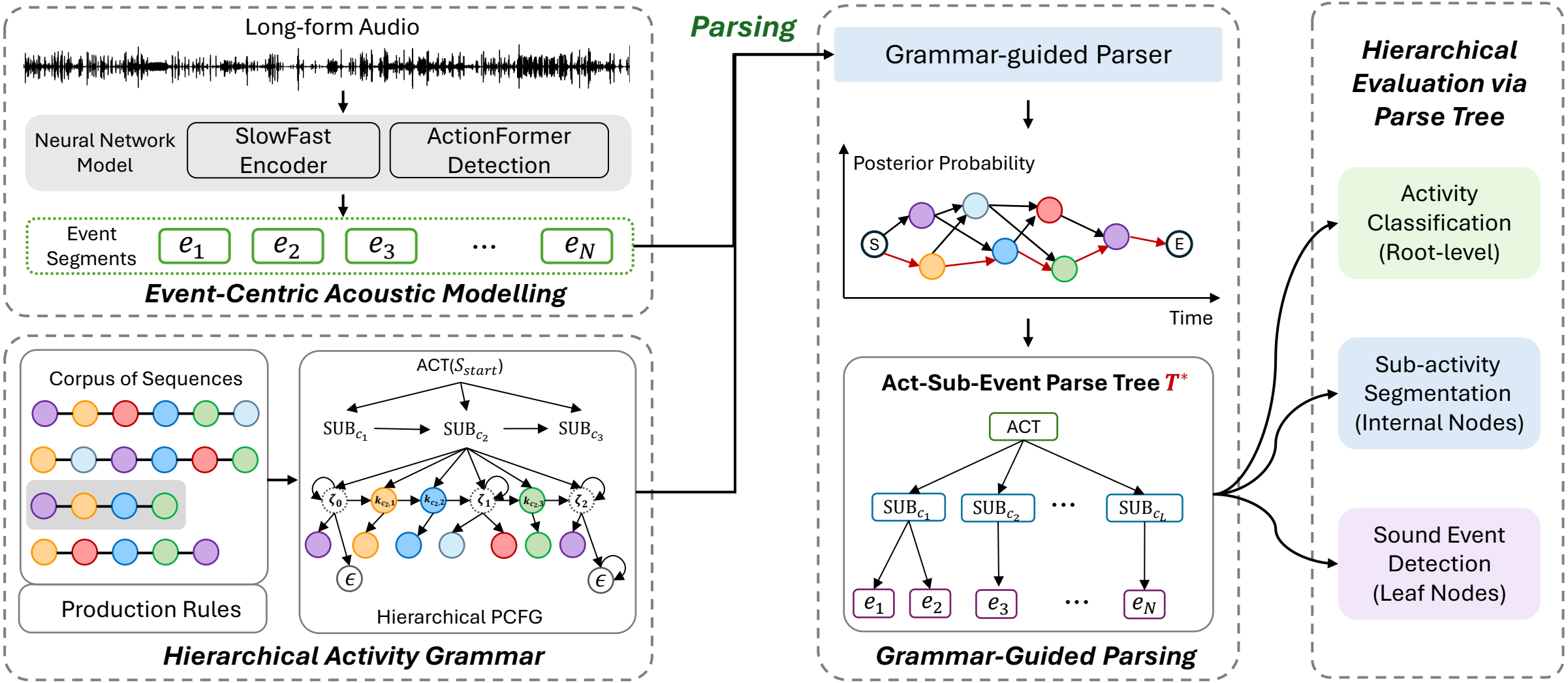


Inspired by the sentence parse tree:





Proposed Grammar-guided Modelling Framework

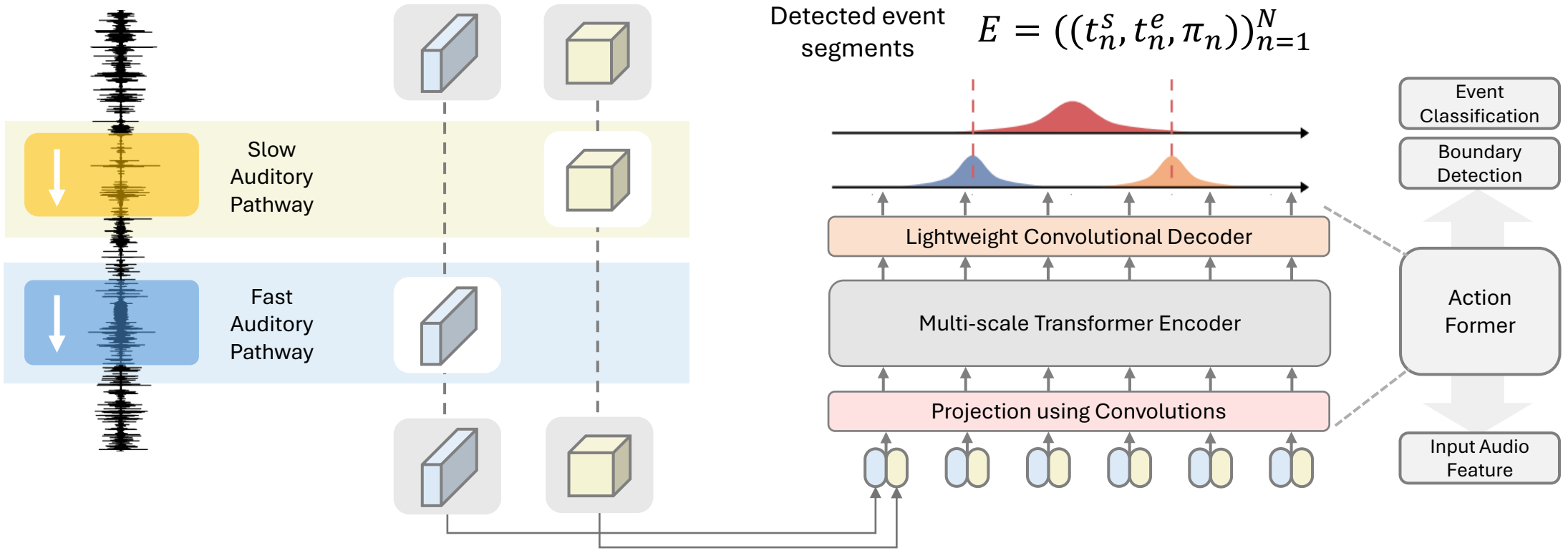


Problem Definition: $T^* = \arg\max_{T \in T(E)} [\log p_{acous}(T|x) + \lambda \log p_{gram}(T)]$

Neural View – Event-centric Acoustic Modelling

From long-form audio to event hypotheses and acoustic likelihoods

1 **Long-form audio** **2** **SlowFast auditory encoder** **3** **ActionFormer temporal modelling**



Symbolic View – Hierarchical Activity Grammar (HAG)

$$\mathcal{G} = (\mathcal{N}, \Sigma, \mathcal{R}, \mathcal{S}_{start}, P)$$

A PCFG defines which hierarchical parses are valid and how likely each parse is.

- **Non-terminal set :**

$$\mathcal{N} = A \cup S \cup N_{noise}$$

- **Terminal event set:**

$$\Sigma = \{water, scrub, collision, \dots\}$$

- **Production rule set:**

Activity-level and sub-activity-level rules

$$\mathcal{R} = \mathcal{R}_{act} \cup \mathcal{R}_{sub}$$

- **Root activity:**

$$\mathcal{S}_{start} \in A$$

- **Production probabilities**

$P(r)$ assigns likelihoods to grammar rules.

For each left-hand side X: $\Sigma_{\alpha(X \to \alpha) \in \mathcal{R}} P(X \to \alpha) = 1$

Production rule r	P(r)
Cleanup → WashScrub WipeTidy	0.45
Cleanup → WashScrub PlateFood WipeTidy	0.30

…

HAG Production Rules

Production rule set: $\mathcal{R} = \mathcal{R}_{act} \cup \mathcal{R}_{sub}$

$\mathcal{R}_{act}$:

Global order of sub-activities inside an activity.

$$\text{ACT} \rightarrow \text{SUB}_{c_1} \ldots \text{SUB}_{c_L}, c_\ell \in \{1, \ldots, |S|\}$$

c_ℓ: sub-activity type at position ℓ.

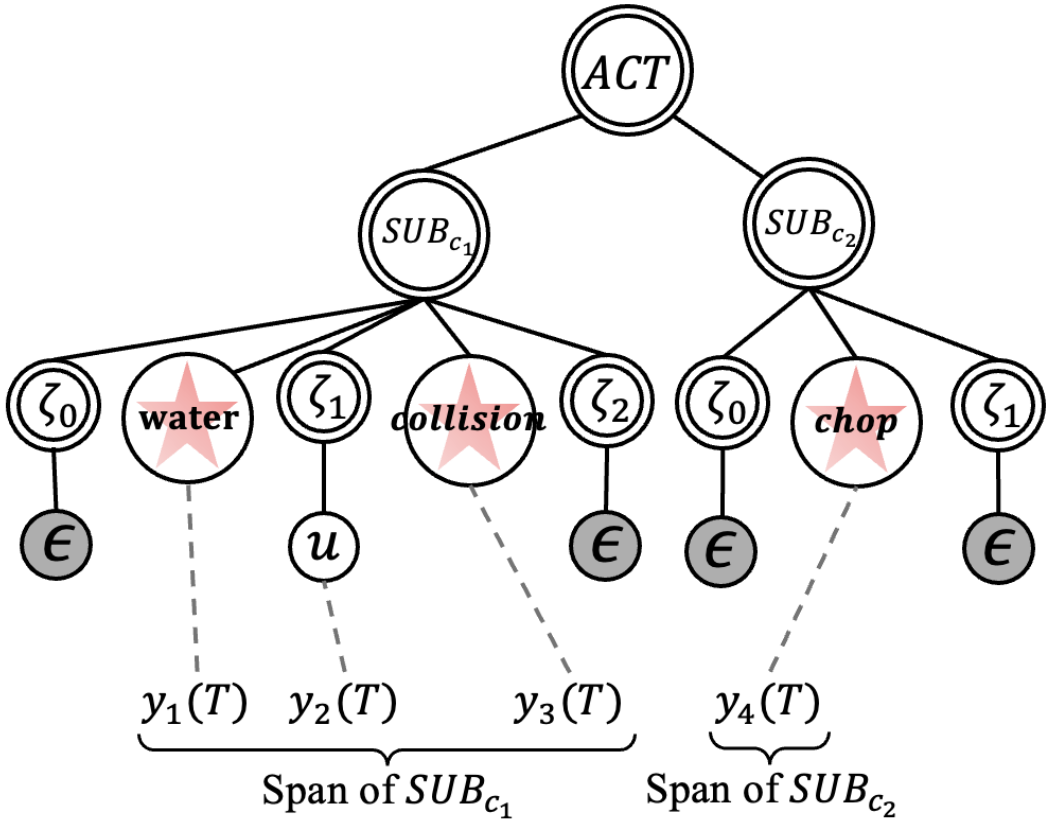
$\mathcal{R}_{sub}$:

Anchors define the expected acoustic backbone;

ζ nodes absorb optional non-anchor events.

$$\text{SUB}_c \rightarrow \zeta_0 k_{c,1} \zeta_1 k_{c,2} \ldots k_{c,M_c} \zeta_{M_c}$$
$$\zeta_i \rightarrow u\, \zeta_i \mid \epsilon, \qquad u \in U_c$$

K_c: anchor set; $U_c = \Sigma \setminus K_c$.



Example Rules:

$ACT \rightarrow SUB_{c_1}\ SUB_{c_2}$

$SUB_{c_1} \rightarrow \zeta_0$ water ζ_1 collision ζ_2

$SUB_{c_2} \rightarrow \zeta_0$ chop ζ_1

Anchor terminal ($\in K_c$)
Non-anchor terminal ($u \in U_c$)
ϵ (Empty)
Non-terminal

Grammar-guided Parsing

Parsing objective:

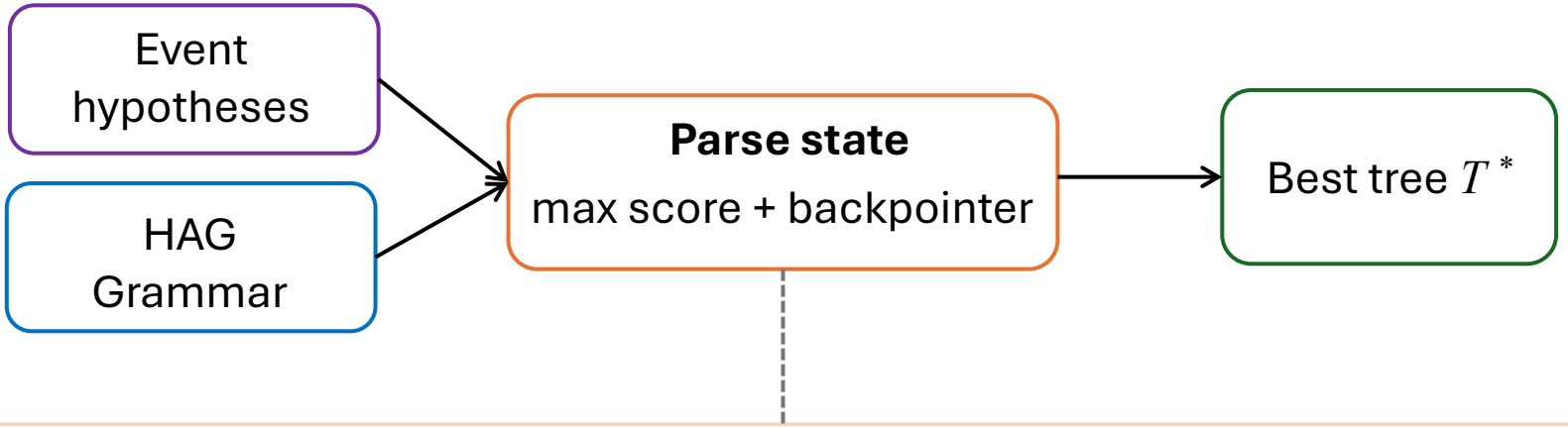
$$T^* = \arg\max_{T \in T(E)} [\log p_{acous}(T|x) + \lambda \log p_{gram}(T)]$$

Acoustic evidence — Grammar prior

λ controls the strength of the grammar prior.

Viterbi-style Earley Parser

Search over valid HAG parse trees and keep the highest-scoring partial derivation.



Experimental Setup

- MultiAct kitchen scenario dataset

Level	#Classes	#Annotations	Duration Range	Avg Duration
Activity	3	51	15.72s ~ 3164.33s	627.90s
Sub-activity	12	472	0.57s ~571.96s	63.56s
Event	44	7312	0.20s ~ 100.02s	4.91s

- Evaluation Protocol: **Event-only training and hierarchical inference**
  - HAG rules/probabilities are induced from training scripts.
  - Grammar is fixed for Val/Eval and used only as a parsing prior.
  - Same event hypotheses support all three hierarchy-level tasks.

Event Detection

AP at IoU thresholds, plus mAP

Sub-activity Segmentation

Edit score, F1@10/25/50, frame-wise accuracy

Activity Classification

Top-1, mPCA, mAP, mAUC

P. Zhang, Q. Luo, P. J. B. Jackson and W. Wang, " Grammar-Guided Hierarchical Parsing for Long-form Audio Activity Recognition," *Interspeech*, Sydney, Australia, 2026.

Event Detection Results

Event detection performance measured by Average Precision (AP) at different temporal IoU thresholds (%, ↑).

Split	Method	AP @ tIoU					mAP
		0.1	0.2	0.3	0.4	0.5	
Val	Event NN†	16.98	14.70	12.81	11.18	9.84	13.10
	Event NN + Grammar‡	**17.00**	**14.79**	**12.83**	**11.19**	**9.85**	**13.13**
Eval	Event NN†	16.48	15.66	14.97	13.55	12.49	14.63
	Event NN + Grammar‡	**16.50**	**15.68**	**14.99**	**13.57**	**12.50**	**14.65**

† Reported in MultiAct [3]. ‡ Evaluated by us with grammar-guided decoding applied after detection.

- Small but consistent AP gain at every tIoU threshold.
- Since parsing is applied after detection, event-level performance is preserved.
- This confirms that HAG parsing does not degrade the acoustic evidence layer.

Sub-activity Segmentation Results

- ***Fully-supervised baseline*** uses sub-activity boundary supervision.
- ***Event-only inference*** uses event predictions as evidence, with no sub-activity labels for training.

Sub-activity segmentation results (%, ↑).

Split	Method	Sup.	Edit	F1@10	F1@25	F1@50	Acc.
Fully-supervised baseline							
Val	Subactivity NN‡	E+S	36.3	39.5	33.5	18.0	38.9
Eval	Subactivity NN‡	E+S	24.6	21.9	17.4	9.0	21.3
Event-only inference							
Val	Grammar-induced (ours)	E	**37.1▲**	24.6	21.1	15.8	25.5
Eval	Grammar-induced (ours)	E	**35.3▲**	**24.8▲**	14.0	7.8	19.7

Sup.: E = event-only; E+S = event + sub-activity boundary supervision. ‡ Evaluated by us using the official MultiAct code/protocol.

- HAG improves global sub-activity order.
- Boundary-sensitive metrics remain limited: parsing cannot refine detector proposal boundaries.

Activity Classification Results

- ***Fully-supervised methods*** uses activity labels.
- ***Event-only grammar induced inference:*** event + grammar → sub-activity → root activity.

High-level activity classification results (%, ↑).

Split	Method	Sup.	Top-1	mPCA	mAP	mAUC
Fully-supervised baseline						
Val	Activity NN†	E+S+A	66.7	61.9	72.7	84.6
Eval	Activity NN†	E+S+A	83.3	83.3	72.2	70.8
Event-only hierarchical inference						
Val	Grammar-induced (ours)	E	**73.3▲**	**70.3▲**	59.5	78.5
Eval	Grammar-induced (ours)	E	66.7	66.7	58.3	**75.0▲**

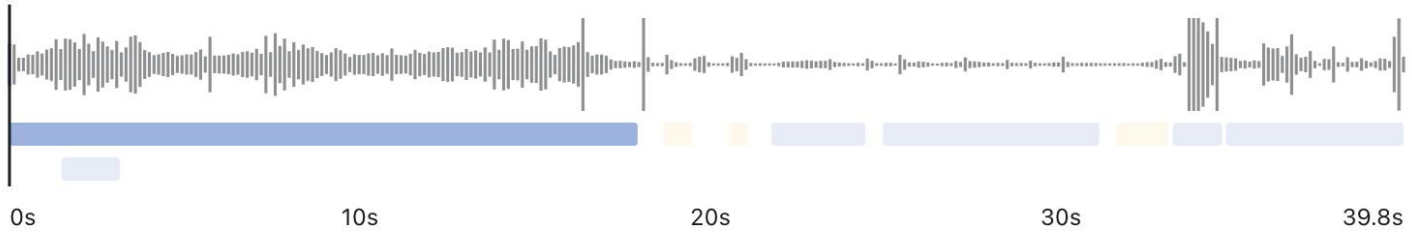
Sup.: E = event-only; E+S+A = event + sub-activity + activity label supervision.
† Reported in MultiAct [3].

- Root activity is inferred from the highest-scoring parse tree.
- No sub-activity or activity labels are used for training.
- Competitive root-level inference, though Eval remains sensitive to imperfect evidence.

Demo



39.78 s



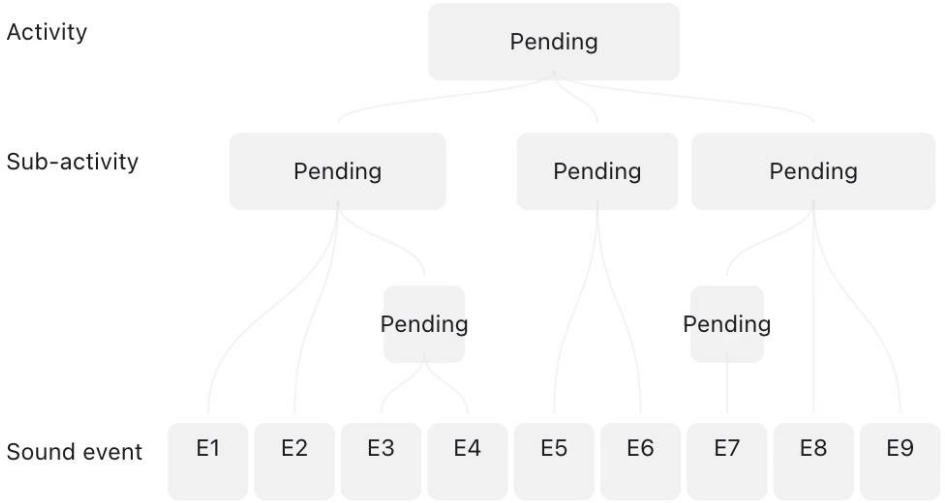
Predict | Scan | Complete

Expand candidate rules

Cleanup → · WashScrub WipeTidy
OpenCloseStorage

Two activity sequences are allowed,
with priors of 0.70 and 0.30.

Hierarchical activity tree



ζ retains optional non-anchor events.

Outline

- Background
- Ego-centric long-form audio activity understanding
  - MultiAct datasets (v1, v2)
  - Hierarchical SlowFast modelling framework
  - Hierarchical grammar-tree parsing
- 360° video to spatial audio generation
  - YT-Speech dataset & a localizer-renderer model
  - AV-QTSS-Syn dataset & AV-QTSS model
- Ego-centric audio-visual understanding
  - A new dataset & benchmark
- Conclusion and future works

Motivation & Background

Why this matters：

- **Immersive 360° media are increasingly used in VR and AR, where spatial audio should remain coherent with the captured visual scene during playback.**
- **Speech-dominant scenes are common in real-world 360° videos, yet spatial audio generation for speech scenes remains relatively underexplored.**



Limitations of existing approaches:

- **End-to-end generation**——requires large-scale training and provides limited control over spatial reconstruction.
- **Two-stage spatial reconstruction**——latent source separation may introduce artifacts and unreliable spatial cues.
- **Zero-shot geometric rendering**——interpretable, but less stable in reverberant or diffuse real-world scenes.

Research Gap:

How can we reconstruct spatial audio for real-world 360° speech videos using reliable and interpretable audio-visual spatial cues?

Q. Luo, P. Zhang, D. Berghi, W. Wang, and P. Jackson, “Visually-Guided Spatial Audio Generation for 360◦ In-the-Wild Speech Scenes,” in Proceedings of Interspeech 2026, Sydney, Australia.

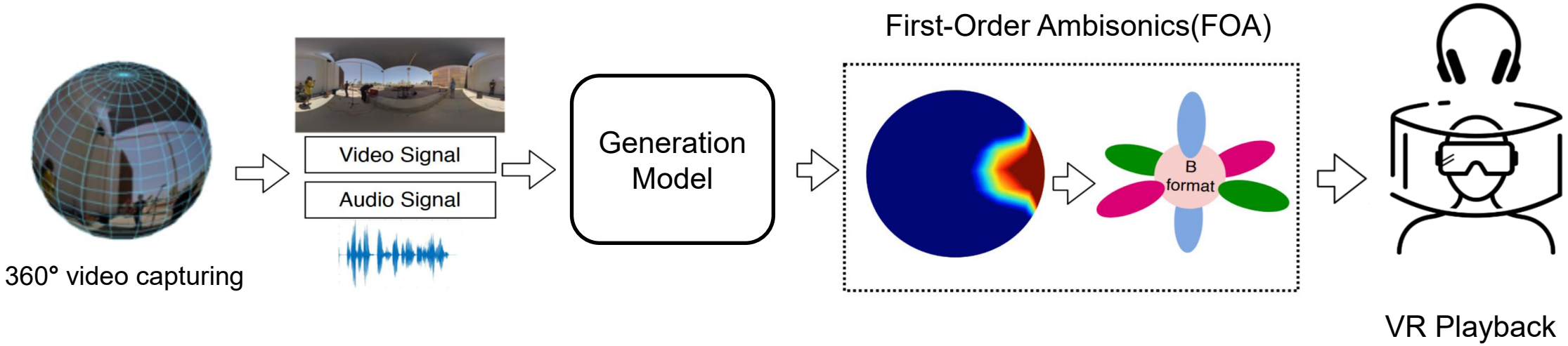
360° Video-Guided Spatial Audio Generation

Goal: Generate spatial audio for ***360-degree videos*** using visual cues.

Spatial audio format: the First-Order Ambisonics (FOA)

- $\mathbf{FOA} = (W, X, Y, Z)$
- W: omnidirectional component
- (X, Y, Z) : directional sound components

Research problem: Given a 360° video and the omnidirectional channel W, predict the missing directional FOA components (X, Y, Z).



Dataset Gap

Tab. Comparison of existing 360°-FOA datasets

Dataset	Length	#Clips	Clips duration	Source	Class
STARSS23 [33]	7.5h	0.2k	<10min	Recorded	13
REC-STREET [9]	3.5h	123k	0.1s	Recorded	1
YT-ALL [9]	113.1h	1.1k	0.1s	YouTube	>8
YT-360 [10]	246h	89k	0.1s	YouTube	32
YT-Ambigen [7]	142h	102k	5s	YouTube	>300
Sphere360 [8]	288h	103k	10s	YouTube	288

Problems:

- Inconsistent FOA channel formats
- Unbalanced channel energy distribution
- No speech-focused filtering

Tab. Comparison of current speech spatial audio datasets.

Dataset	Hours	Audio Format	Source	Labels	Audio type
EasyCom [34]	5	Binaural	Recorded	Text, Geometric	speech
Binaural [35]	2	Binaural	Recorded	Geometric	speech
MRSDrama [36]	98	Binaural	Recorded	Text, Video, Geometric	speech
Spatial LibriSpeech [37]	650	FOA	Simulated	Text, Geometric	speech
YT-SPEECH	8.9	FOA	Youtube	360°Video	speech

Problems:

- Limited spatial formats, mostly binaural
- Limited real-world visual grounding

Proposed Dataset: YT-SPEECH Overview & Curation Pipeline

Definition:

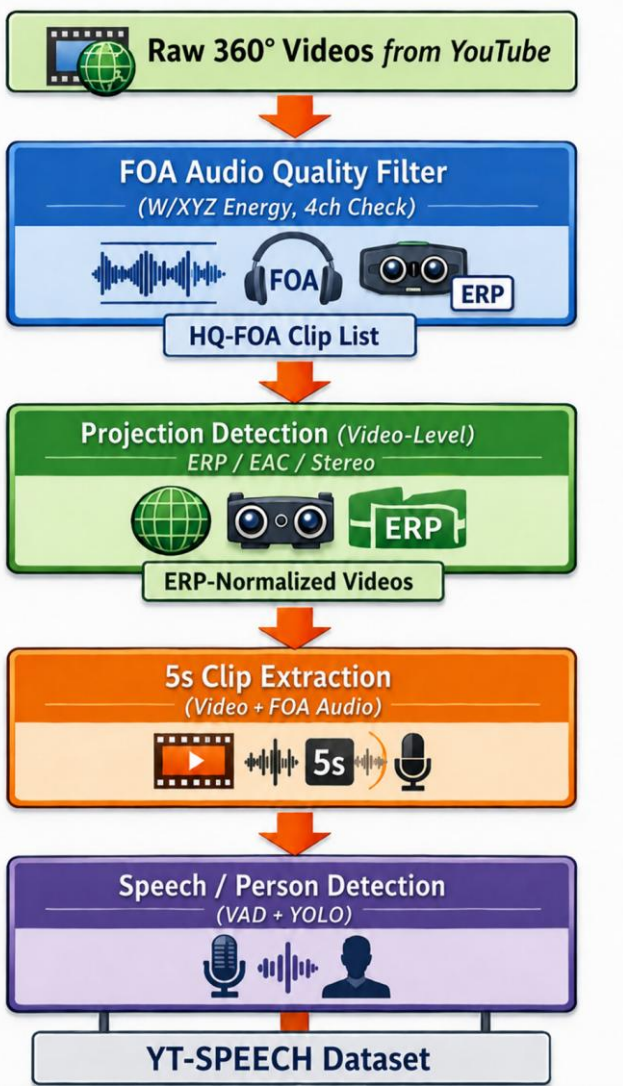
A **speech-oriented, high-quality, real-world** 360° video–FOA dataset for video-guided FOA reconstruction.

Dataset Statistics:

- **8.9 hours** of curated FOA speech recordings
- 5-second clips at 24 kHz
- **197 source videos**
- Train / Val / Test split by video IDs

Dataset Characteristics:

- Real-world **360° visual–audio pairs**
- **Speech-dominant scenes** with reliable FOA spatial cues
- Diverse visual and acoustic conditions



Example Scenes from YT-SPEECH

YT SPEECH covers diverse real-world speech scenarios: indoor and outdoor environments, solo and group interactions, near- and far-field speech, and quiet/noisy background conditions.



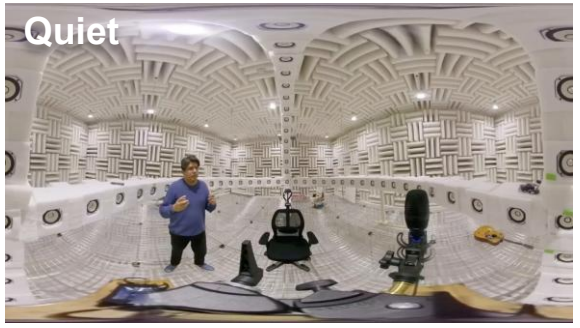














Q. Luo, P. Zhang, D. Berghi, W. Wang, and P. Jackson, “Visually-Guided Spatial Audio Generation for 360◦ In-the-Wild Speech Scenes,” in Proceedings of Interspeech 2026, Sydney, Australia.

A Localizer–Renderer Model

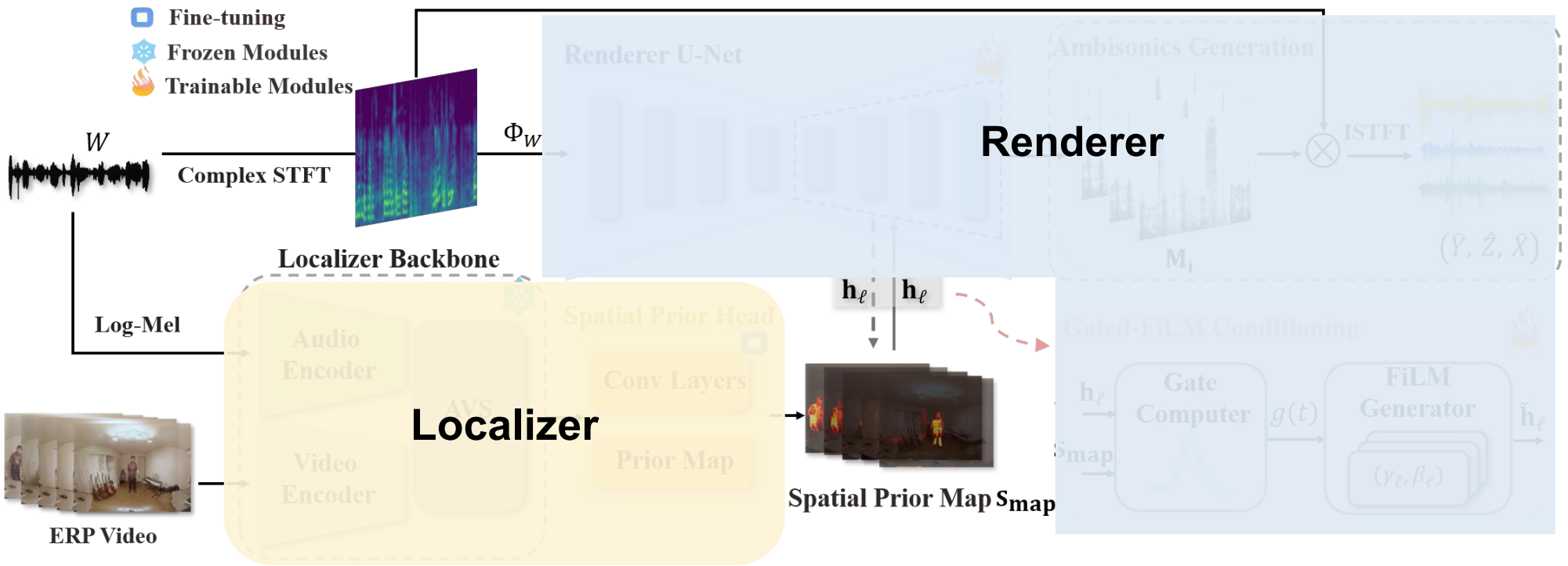
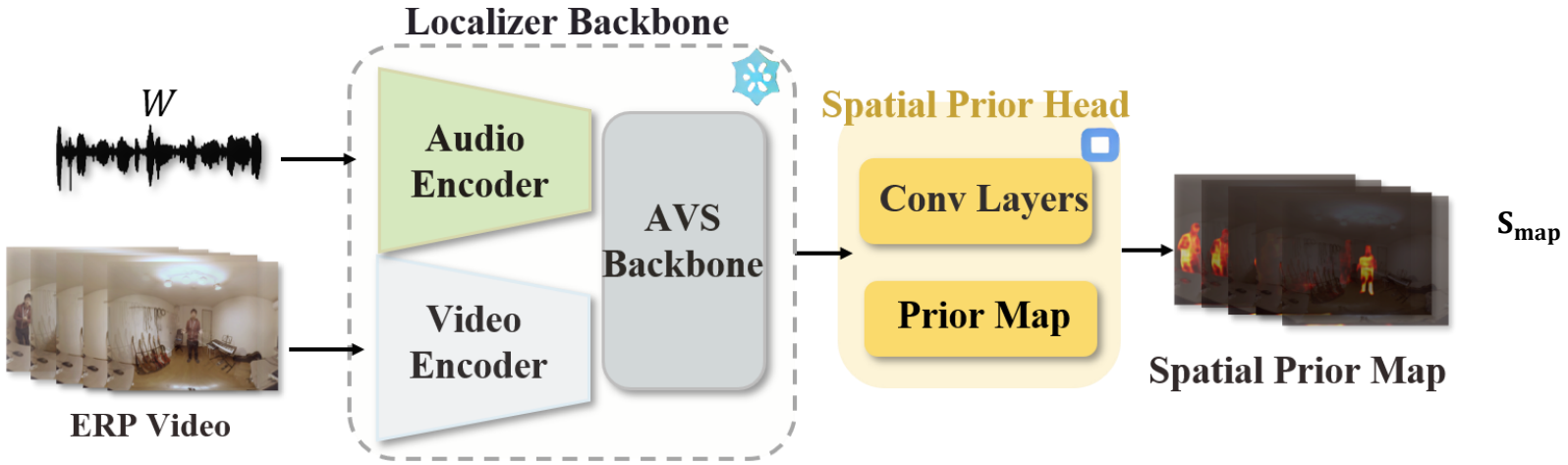


Fig. Overview of the proposed Localizer–Renderer framework for visually guided FOA spatial reconstruction.

- **Stage-1 (Localizer):** Estimates dense audio-visual spatial priors from ERP video and W-channel audio.
- **Stage-2 (Renderer):** Uses the spatial prior to reconstruct directional FOA components X, Y, Z from W.

◆ Model Details - Localizer



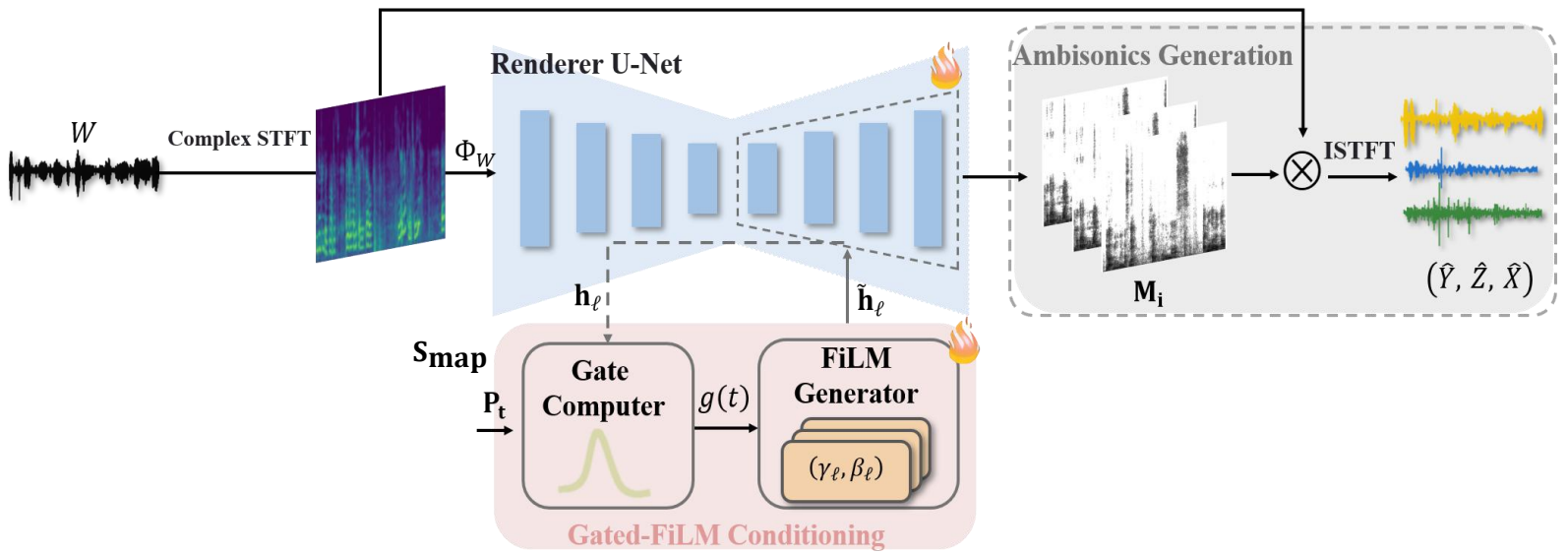
Goal: Estimate a dense spatial prior map over the 360° scene.

Inputs : ERP video frame $\mathbf{V} \in \mathbb{R}^{3\times224\times448}$ + omnidirectional audio W

Backbone:
- Frozen AVS backbone with audio encoders
- The spatial prior head predicts sounding-object heatmaps

Outputs: Normalized spatial prior map $\mathbf{S}_{\text{map}} \in \mathbb{R}^{7\times14}$ with $\sum_i \mathbf{S}_{\mathbf{map},i} = 1$

◆ Model Details - Renderer



Goal: Reconstruct directional FOA spectra from W using the spatial prior.
Inputs : Complex STFT of W + spatial prior map
Renderer :

- A complex-domain U-Net transforms the omnidirectional spectrum into directional components.

Gated FiLM :

- Confidence gate controls how strongly visual spatial priors condition the Renderer
- Reduces unreliable visual conditioning in ambiguous scenes

$$\tilde{\mathbf{h}}_\ell = \mathbf{h}_\ell \odot (1 + g(t)\boldsymbol{\gamma}_\ell) + g(t)\boldsymbol{\beta}_\ell$$

Outputs: reconstructed directional FOA channels X, Y, Z

Experiments Setup

◆ Training strategy:

A. Renderer pre-training on a large-scale 360° audio-visual dataset

- Localizer frozen
- Renderer learns prior-conditioned FOA reconstruction.

B. Speech-oriented fine-tuning on YT-SPEECH dataset

- Prior Head and Renderer jointly optimized
- Adapts the model to person-aware speech spatialization

◆ Data Augmentation:

- Random yaw rotation with corresponding FOA channel rotation
- Horizontal flips for rotational robustness

Q. Luo, P. Zhang, D. Berghi, W. Wang, and P. Jackson, “Visually-Guided Spatial Audio Generation for 360◦ In-the-Wild Speech Scenes,” in Proceedings of Interspeech 2026, Sydney, Australia.

◆ Quantitative Results on YT-SPEECH

Localizer ablation and Renderer comparison under the Localizer–Renderer framework on YTSPEECH dataset. The best score for each metric is marked in bold and the second best score is underlined.

Model	Reconstruction Metrics				Spatial Metrics			Speech Metrics		
	$\ell_2(\times 10^3)\downarrow$	$\mathcal{L}_{\text{mag}}\downarrow$	$\mathcal{L}_{\text{phs}}\downarrow$	$\mathcal{L}_{\text{MRS}}\downarrow$	$\Delta_{\text{abs}}\theta\downarrow$	$\Delta_{\text{abs}}\phi\downarrow$	$\Delta_{\text{ang}}\downarrow$	PESQ ↑	MOS-Q ↑	MOS-P ↑
Ablation on Localizer Components										
NoVideo-Renderer	1.59	0.55	1.96	2.33	0.69	0.30	0.73	2.50	3.24	2.28
VidEnc-Renderer	1.95	0.50	1.66	<u>2.11</u>	<u>0.59</u>	<u>0.23</u>	<u>0.66</u>	1.78	3.03	<u>3.02</u>
Loc(frz)-Renderer	2.19	0.53	1.90	2.58	0.97	0.44	1.04	1.41	2.49	2.34
Comparison to Renderer Methods										
Localizer-AmbiEnc	<u>1.55</u>	**0.34**	1.67	**1.69**	0.61	0.50	0.83	<u>3.35</u>	**3.68**	2.97
Localizer-Pyroom	2.22	0.58	1.82	2.56	**0.58**	0.50	0.83	1.69	2.28	2.51
Ours (NoPT)	1.71	0.52	**1.59**	2.46	0.84	0.36	0.92	1.48	2.94	2.90
Ours	**1.15**	<u>0.44</u>	**1.59**	2.41	0.60	**0.18**	**0.64**	**3.42**	<u>3.63</u>	**3.16**

Key observations:

- The Localizer significantly improves spatial accuracy, reducing the angular error Δang
- The full model achieves the best reconstruction error ℓ_2 and strong spatial performance.
- The learned renderer also improves perceptual speech quality, achieving the best PESQ and MOS-P.

◆ Cross-Dataset Compatibility

Compatibility comparison with SAG on YT-ALL, YT-MUSIC, YT-CLEAN and YT-SPEECH datasets

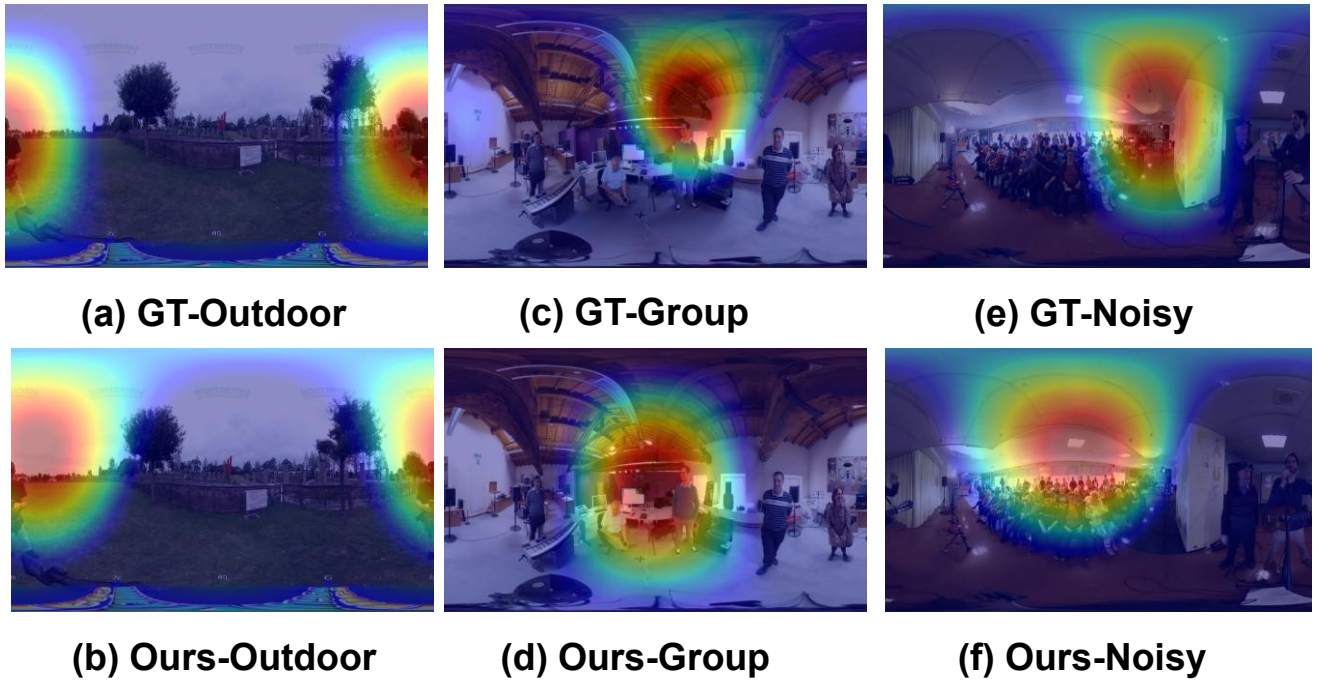
	YT-ALL		YT-MUSIC		YT-CLEAN		YT-SPEECH	
	STFT↓	ENV↓	STFT↓	ENV↓	STFT↓	ENV↓	STFT↓	ENV↓
SAG(NoVideo)	2.96	3.48	4.55	4.32	1.57	2.12	1.43	2.42
SAG	**2.78**	**2.92**	4.46	3.77	1.58	1.80	1.14	2.18
Ours(NoVideo)	3.12	3.52	3.98	4.01	1.07	1.78	1.49	1.89
Ours	2.91	2.99	**3.38**	**3.17**	**0.91**	**1.78**	**0.85**	**1.71**

Key observations:

- Clear gains on **YT-CLEAN and YT-SPEECH**, where sources are visually localizable.
- On **YT-MUSIC**, our method improves both STFT and ENV despite mixed sources.
- On **YT-ALL**, gains are smaller due to more diverse non-speech content.

◆ Qualitative Results

Spatial Audio Energy Map Visualization on the YT-SPEECH Test Set:



(a) GT-Outdoor **(c) GT-Group** **(e) GT-Noisy**

(b) Ours-Outdoor **(d) Ours-Group** **(f) Ours-Noisy**

Columns I–III show outdoor, group, and noisy scenes. Ground truth and our prediction are overlaid on the 360◦ frames in (a,c,e) and (b,d,f), respectively.

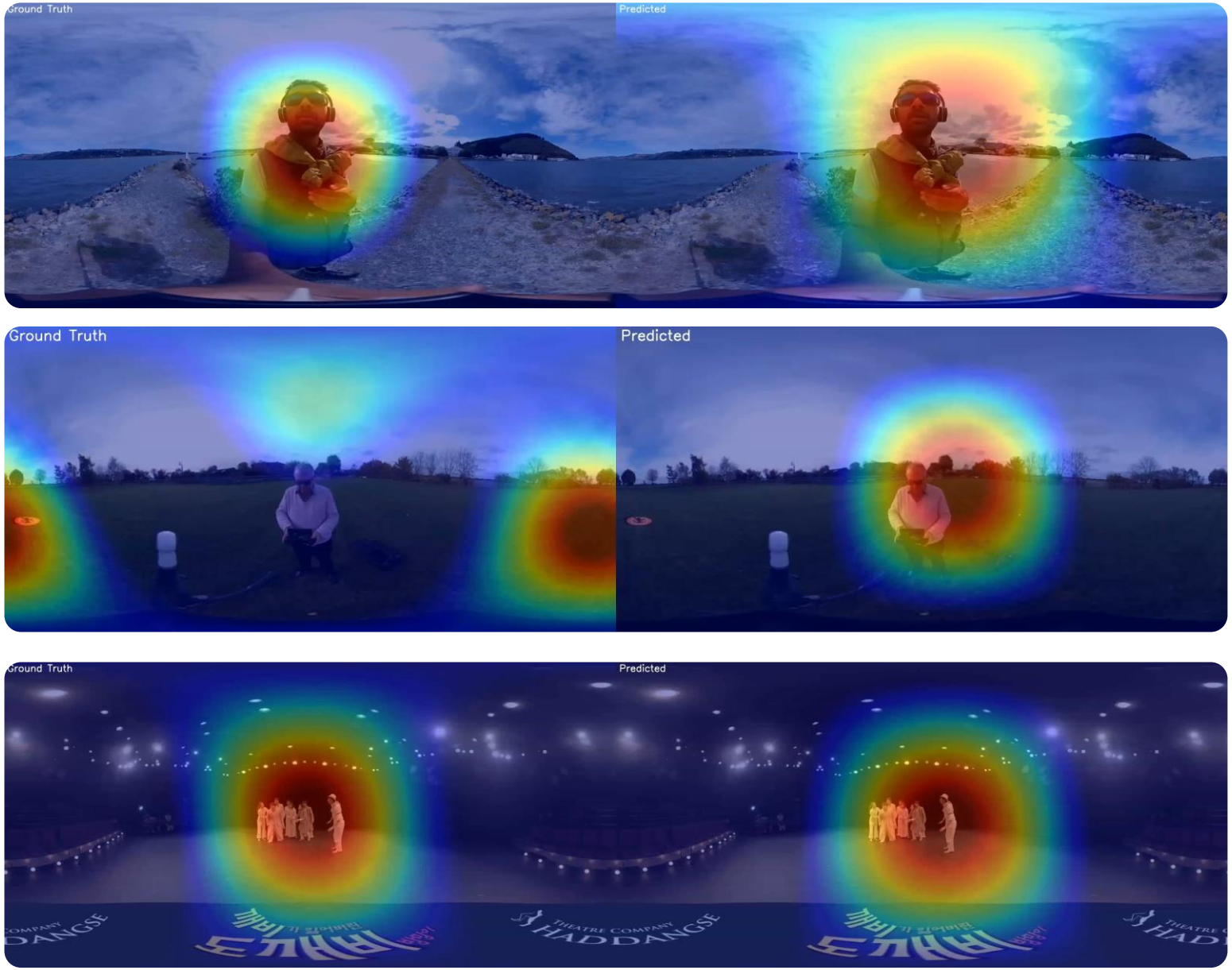
Results:

- Predictions are generally aligned with visually observable sound sources.
- Noisy backgrounds can introduce spurious responses, indicating a remaining failure mode.

Demos

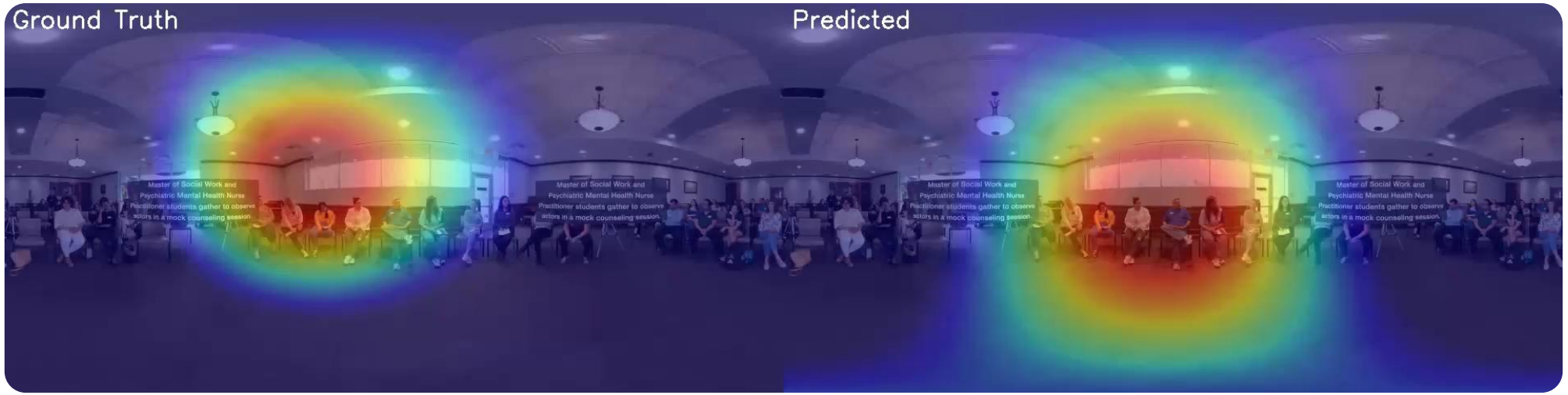
Ground-Truth

Prediction(Ours)



Demos (Failed Cases)

- **Multi-Person Scenes：** The model may struggle to identify the active speaker when multiple visible speakers are present.



- **Overlapping-Sources Scenes:** The model does not explicitly disentangle multiple simultaneous sources and may focus on a dominant direction.



Motivation

- **360° media editing and playback** often **need control over one source**: — isolating a speaker, enhancing it, suppressing a distracting noise, or moving it to another direction.
- **Existing recordings mixes all sources**, so no source is directly addressable. Audio-visual spatialization (incl. our first task) and video-to-FOA generation **typically output a whole-scene soundtrack.**
- Selecting a source needs both its sound and its position, but current **separation models return the sound without its position**, while current **localization models find sounding objects without extracting their sound**.

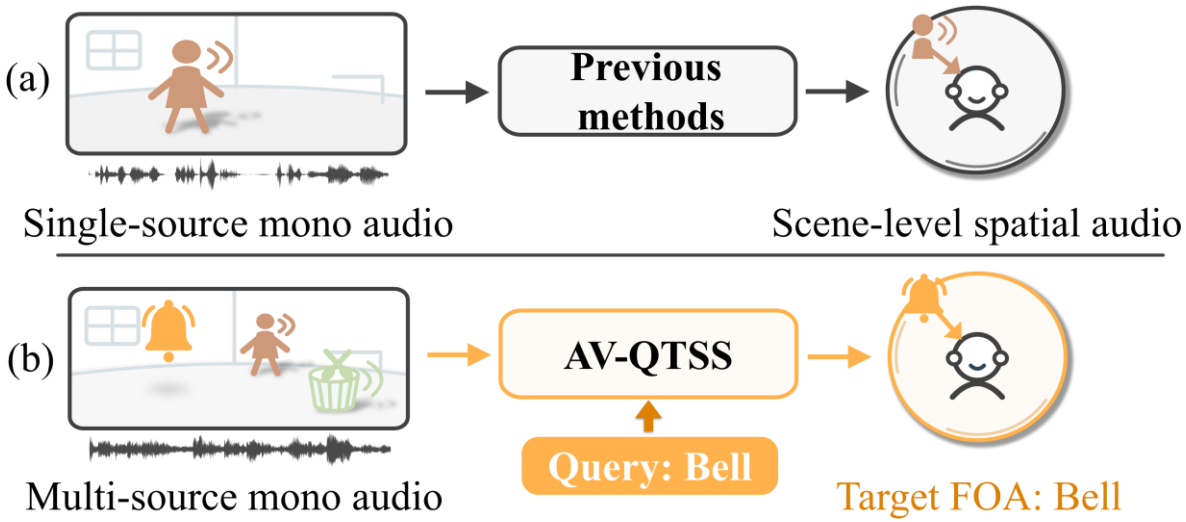


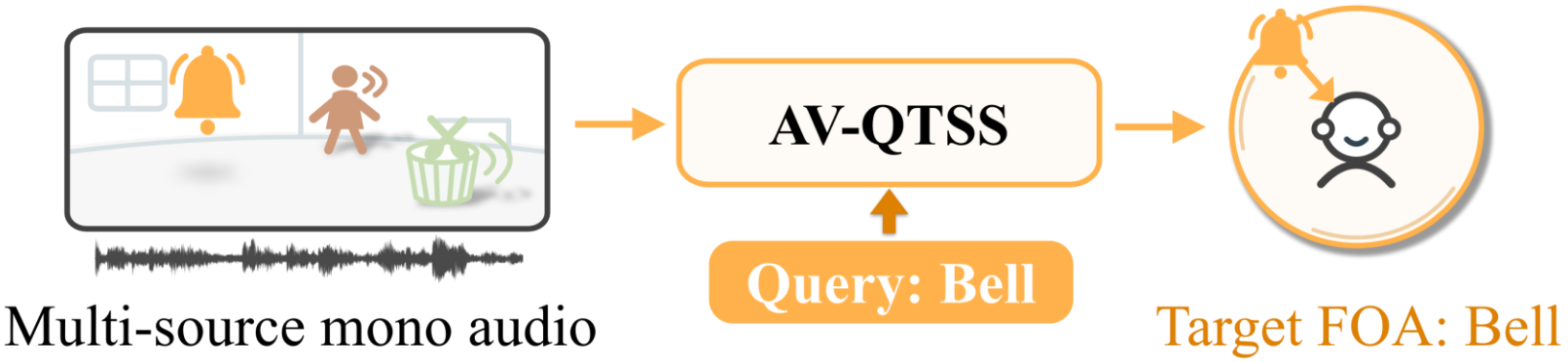
Fig. (a) Scene-level spatialization. (b) Class-query target spatialization

Audio-Visual Class-Query Target Sound Spatialization (AV-QTSS)

Given a **mono mixture** W_{mix} of overlapping sources, a **360° video V** and a **class query** q, AV-QTSS outputs the **FOA of the queried source only**, with omnidirectional channel $\hat{W}_q$ and directional channels $\hat{Y}_q$, $\hat{Z}_q$, $\hat{X}_q$:

$$\mathcal{F}: (W_{\text{mix}}, \mathbf{V}, q) \mapsto \hat{\mathbf{F}}_q = [\hat{W}_q, \hat{Y}_q, \hat{Z}_q, \hat{X}_q]$$

Since W_{mix} has no spatial cues, the direction must come from $\mathbf{V}$: the same q must extract the sound and find its visual source.



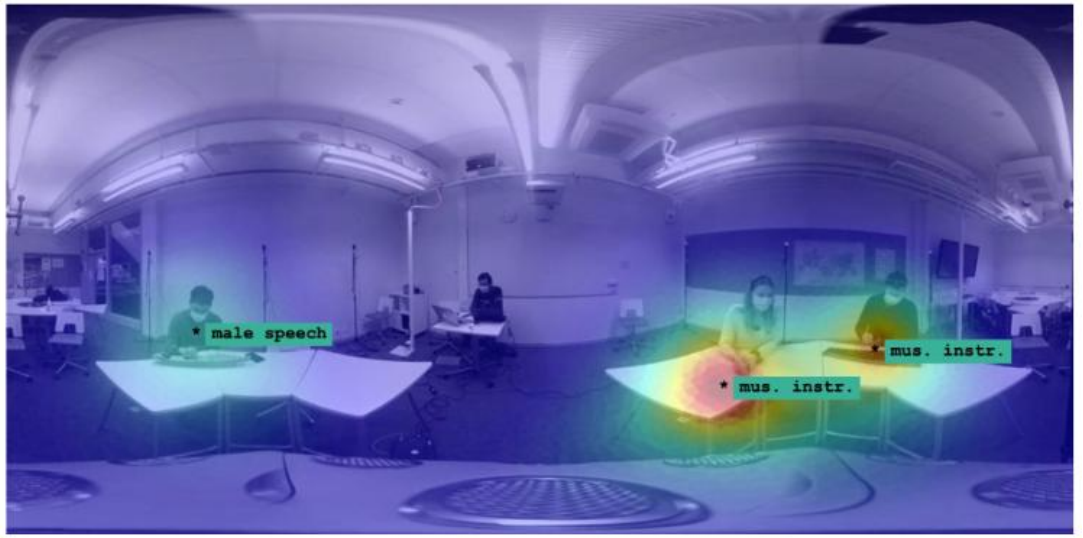
Q. Luo, P. Zhang, D. Berghi, W. Wang, and P. Jackson, "AV-QTSS: Class-query target sound spatialization in multi-source 360° videos," IEEE ICASSP 2027, submitted.

Real Dataset: STAIRS26

What STAIRS26 provides

- **7.5 h** of real 360° recordings;
- 4-channel **mixture FOA** + 360° video with acoustic imaging maps;
- Scenes **contain up to four sources** at the same time, and every frame is labelled with the sound classes (13 DCASE sound event classes).

Acoustic imaging maps



0. **Female speech, woman speaking**
1. **Male speech, man speaking**
2. **Clapping**
3. **Telephone**
4. **Laughter**
5. **Domestic sounds**
6. **Walk, footsteps**
7. **Door, open or close**
8. **Music**
9. **Musical instrument**
10. **Water tap, faucet**
11. **Bell**
12. **Knock**

What it lacks

- **No per-source waveforms or FOA** → it supervises only **real-domain localization**

⇨ **Our split: 156 recordings with 360° video, room-disjoint 94 / 27 / 35**

[1] Iran R. Roman et al., "STAIRS26: Sony-Tau acoustic images of real-world scapes 2026," Zenodo, 2026

Synthetic Dataset: AV-QTSS-Syn

- **80 h of 5-s clips with per-source supervision**, built with SpatialScaper [1] + SELDVisualSynth [2] under the same 13-class ontology as STAIRS26

	Audio-only portion	AV-complete portion
Per clip	Mono mixture + per-source W (reverberant)	**+** 360° video, per-source FOA, direction trajectories
Overlap	50% full / 50% partial (all sources active ≥ 0.5 s)	All sources active for the full 5 s
Used for	Separator adaptation	Localizer & Renderer; all synthetic results are reported on its test set.
Clips (train / val / test)	24,000 / 2,400 / 2,400 (40 h)	24,000 / 2,400 / 2,400 (40 h)

➢ Scene Settings

- 24 kHz, 1920×960 ERP video
- **1–4 concurrent** sources (K = 1/2/3/4: 10/40/35/15%)
- **Static and moving** sources: all-static / mixed / all-moving = 25 / 50 / 25% for K ≥ 2
- **Ambience** in half of the clips at 5–20 dB SNR

[1] Iran R. Roman et al., "SpatialScaper: A library to simulate and augment soundscapes for sound event localization and detection in realistic rooms," in Proc. ICASSP, 2024.
[2] Adrian S. Roman et al., "Generating diverse audio-visual 360◦soundscapes for sound event localization and detection," arXivpreprint arXiv:2504.02988, 2025.

Demos of AV-QTSS-Syn

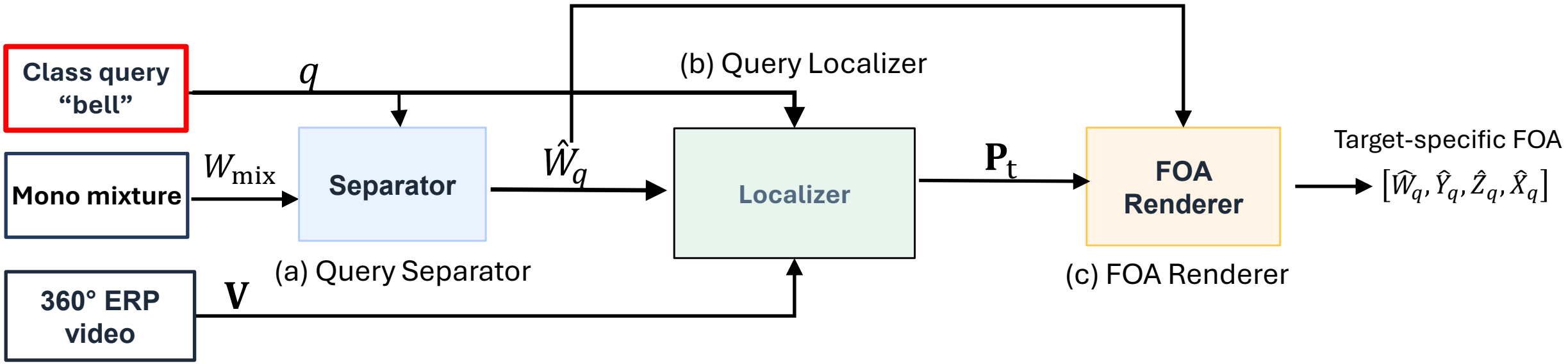
- **Two moving sources** (Male speech, Bell)



- **Three moving source** (Cupboard, Telephone, Water tap)



AV-QTSS: A Separator-Localizer-Render Model



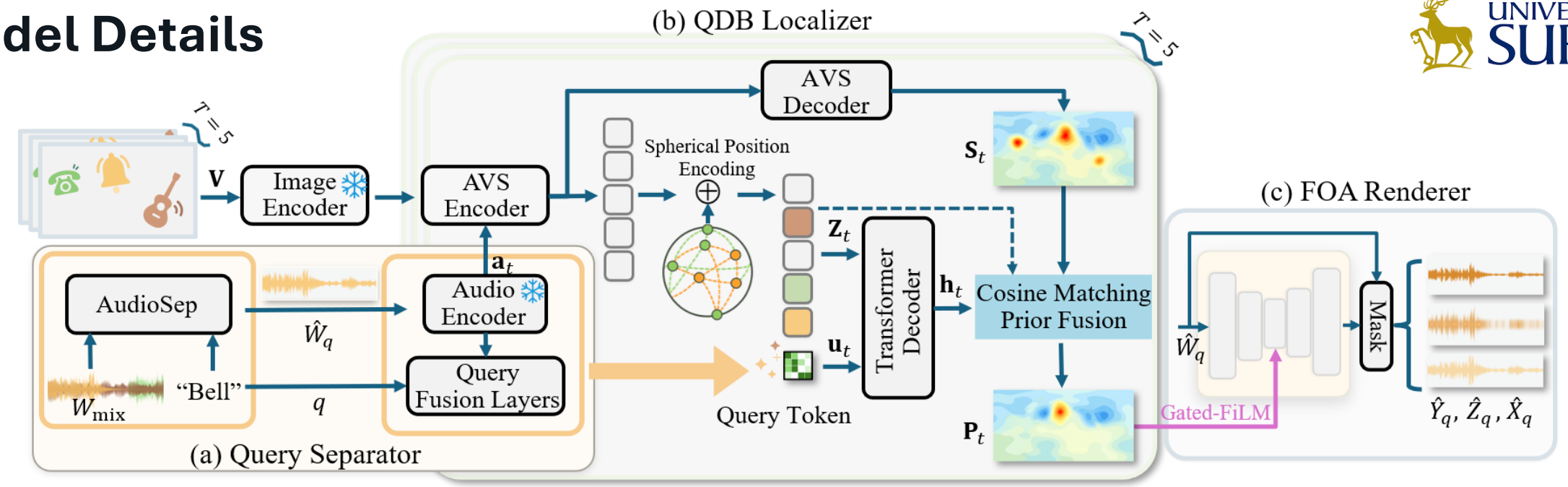
(a) Query Separator—— what the target sounds like: $(W_{\text{mix}}, q) \rightarrow \hat{W}_q$

(b) Query Localizer —— where it is: $(V, \hat{W}_q, q) \rightarrow \mathbf{P}_t$

(c) FOA Renderer(first work style): how it is spatialized $(\hat{W}_q, \mathbf{P}_t) \rightarrow [\hat{W}_q, \hat{Y}_q, \hat{Z}_q, \hat{X}_q]$

Q. Luo, P. Zhang, D. Berghi, W. Wang, and P. Jackson, "AV-QTSS: Class-query target sound spatialization in multi-source 360° videos," IEEE ICASSP 2027, submitted.

Model Details



a. **Query Separator**: we **use AudioSep,** a text-queried separation model fine-tuned on our 13 classes. From the monaural mixture it extracts the reverberant W channel of the queried source.

b. **QDB (Query-Conditioned Dense Binding) Localizer:** built on **the AVS model**, which gives a sounding-region map $\mathbf{S}_t$ but no class identity. We turn the scene into $\mathbf{Z}_t$, one token per spherical cell carrying its audio-visual feature and its direction. We additionally takes class query fused with separated audio into one query token. A **Transformer decoder** takes this query and the whole scene and returns one target-matching vector $\mathbf{h}_t$. Its cosine similarity with every cell of $\mathbf{Z}_t$ scores how well each cell matches the query, and fusing these scores gives $\mathbf{P}_t$, a map that peaks on the queried source.

c. **FOA Renderer:** our first task's U-Net with gated FiLM, conditioned on the map $\mathbf{P}_t$, predicts a complex mask per channel and turns $\hat{W}_q$ into the three directional channels.

Training Strategy

STEP 1: Separator Adaptation (AV-QTSS-Syn, audio-only)

- **Goal:** Extract the reverberant W of the queried class from 1–4-source mixtures.
- **Details:** Staged fine-tuning AudioSep with different learning rates

STEP 2: Joint Optimization (AV-QTSS-Syn, AV-complete)

- **Goal:** Render FOA from the map, so the rendering error also refines the Localizer.
- **Details:** QDB decoder and Renderer are trained jointly; the Separator and the AVS backbone are frozen.

STEP 3: Real-Domain Adaptation (STAIRS26)

- **Goal:** Transfer query localization to real recordings.
- **Details:** Only the QDB decoder is updated; without per-source references, the Renderer is not trained.

Q. Luo, P. Zhang, D. Berghi, W. Wang, and P. Jackson, “AV-QTSS: Class-query target sound spatialization in multi-source 360° videos,” IEEE ICASSP 2027, submitted.

Experiments

Stage-wise Comparison and Ablation on AV-QTSS-Syn

- No existing method shares the complete AV-QTSS setting, we therefore evaluate task-compatible component baselines and controlled system variants under fixed interfaces.
- **USS :** another separator, same class condition; **No-Binding**: original AVS localizer model; **No-Class**: QDB localizer without the class embedding; **Ambi-Encode:** learned Renderer replaced by analytic FOA encoding; **Direction-Only**: Renderer conditioned on peak direction only.

Model	Reconstruction Metrics				Spatial Metrics			Subjective Metrics	
	$\ell_2(\times 10^3)\downarrow$	$\mathcal{L}_{\text{mag}}\downarrow$	$\mathcal{L}_{\text{phs}}\downarrow$	$\mathcal{L}_{\text{MRS}}\downarrow$	$\Delta_{\text{abs}}\theta\downarrow$	$\Delta_{\text{abs}}\phi\downarrow$	$\Delta_{\text{ang}}\downarrow$	MOS-C ↑	MOS-S ↑
Comparison to Separator Methods									
No-Separator	21.7	1.50	1.65	4.70	29.5°	19.1°	39.2°	2.33	2.95
USS [35]	3.66	0.82	1.64	3.43	19.4°	12.4°	26.4°	3.05	3.36
Ablation on Localizer Components									
No-Binding [12]	4.07	0.62	1.70	3.20	43.9°	13.5°	47.4°	3.85	2.44
No-Class	3.83	0.59	1.67	3.18	34.3°	13.2°	38.7°	4.04	2.80
Comparison to Renderer Methods									
Ambi-Encode [1]	3.53	0.60	**1.63**	3.50	**17.3°**	12.6°	25.5°	3.75	3.73
Direction-Only	3.10	**0.57**	1.64	3.14	18.4°	12.1°	24.4°	3.84	4.05
Ours (No-Joint)	3.49	0.61	1.66	3.12	23.4°	**11.5°**	27.7°	4.20	3.87
Ours	**3.01**	0.58	**1.63**	**3.11**	17.4°	12.3°	**23.3°**	**4.24**	**4.15**

Metrics:

Reconstruction on Y, Z, X:
ℓ2 (waveform MSE),
L_mag / L_phs (magnitude / phase RMSE),
L_MRS (multi-resolution STFT distance)

Spatial:
azimuth, elevation and overall angle errors

Subjective:
MOS-C (content)
MOS-S (spatial),

- Binding the query to the right object is doing most of the spatial work in this system.
- Our full method gives the best balance across reconstruction metrics.

Demos-AV-QTSS-Syn

- Each video shows four clips in sequence: the mixture audio , the ground-truth target audio, our predicted target audio, and the predicted target audio with its FOA energy heatmap overlaid.

- All the audio you hear has been binaurally decoded from FOA for headphone playback.

- **Query “Water tap”**



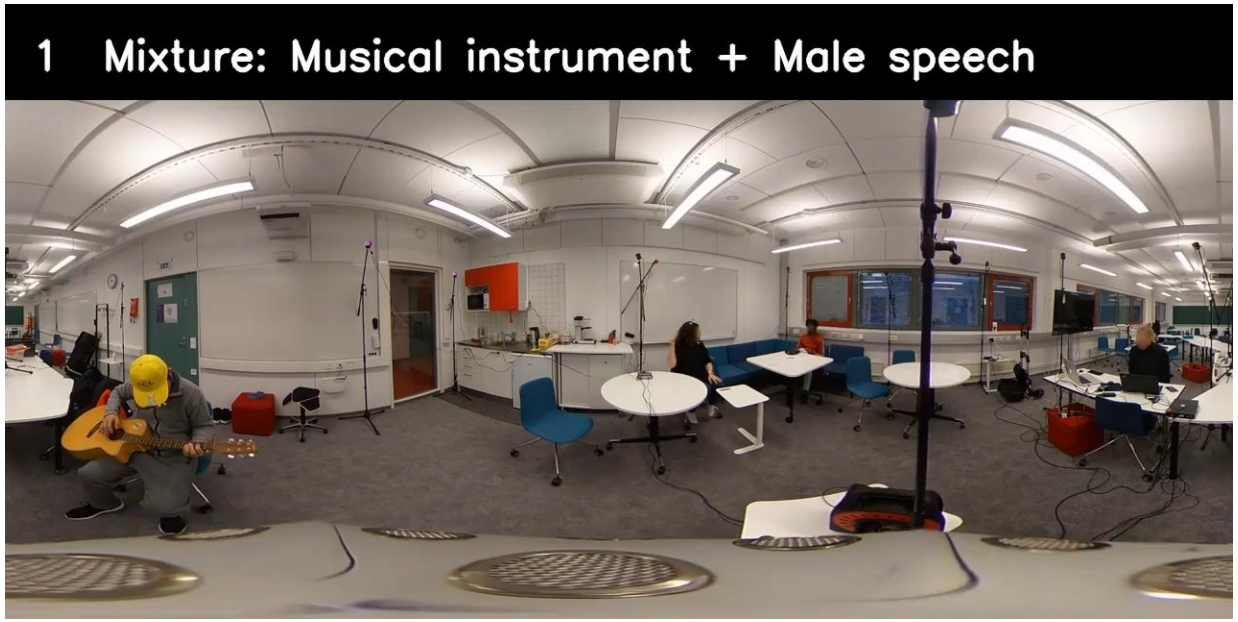
- **Query “bell”**



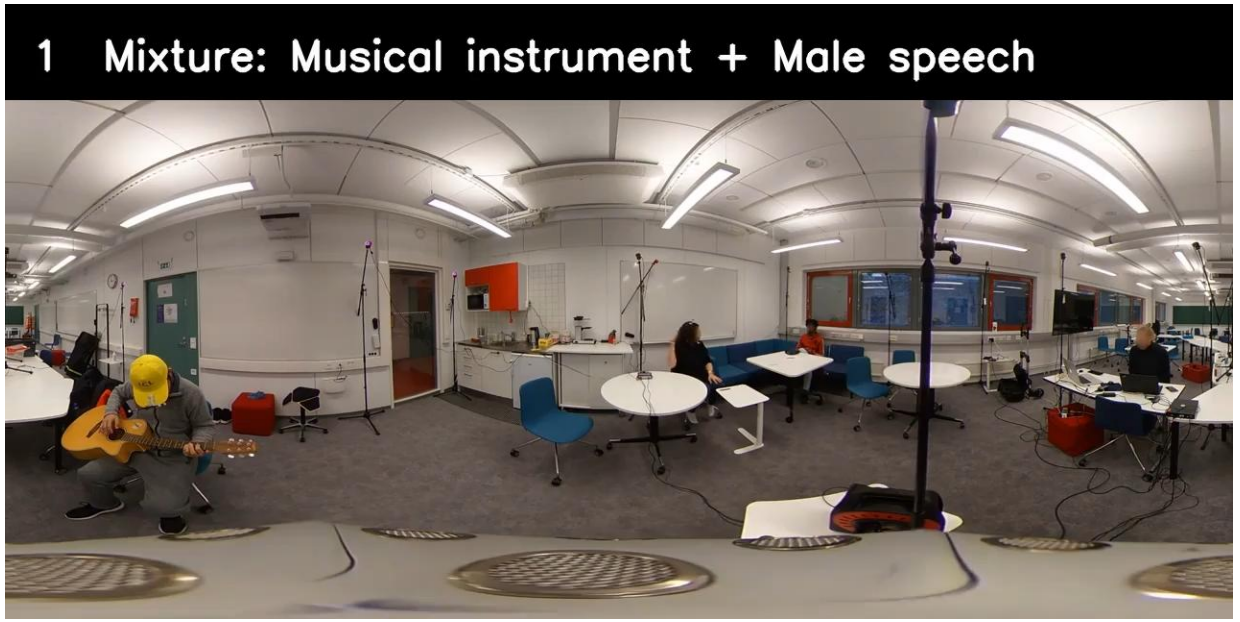
Demos-STAIRS26

Same structure as before, except **STAIRS26 has no per-source ground truth audio**, so each video only has three clips: mixture audio, predicted target audio, and predicted target audio with the heatmap.

- **Query “Musical instrument”**



- **Query “Male speech”**



- **LIMITATION:** our class query identifies what *kind of sound to extract*, not *which individual source*. when several sources share the same- class/same-semantic , our model cannot distinguish them.

Outline

- Background
- Ego-centric long-form audio activity understanding
  - MultiAct datasets (v1, v2)
  - Hierarchical SlowFast modelling framework
  - Hierarchical grammar-tree parsing
- 360° video to spatial audio generation
  - YT-Speech dataset & a localizer-renderer model
  - AV-QTSS-Syn dataset & AV-QTSS model
- Ego-centric audio-visual understanding
  - A new dataset & benchmark
- Conclusion and future works

Understanding Relations Between Actions and Sounds/Vision via World Models

Predict what we will see and hear after an action



Example: starting a washing machine

Vision: the start button is pressed
→ but has it actually started?

Sound: the motor starts, water flows in
→ “Okay, it’s working.”

Sound helps us understand what we cannot see, and can help a world model in the same way

Related Work and Motivation

UNIVERSITY OF SURREY

Visual world models

Well established over several years

e.g. DreamerV3, V-JEPA 2, Genie

Joint audio-video generators

Pretrained backbones to generate future observations

e.g. LTX-2, Ovi

Audio-visual world models **Recent**

⚠ **However:** reliability and generalization to everyday interactions remain limited. One possible reason is the training data.

Model	Training data	Control
AV-CDiT	AVW-4k: simulated indoor navigation	Navigation actions
EchoWM	Gameplay, engine simulation, web video	6-DoF camera trajectories
HelixWorld	Real-world exploration, game footage	Camera paths, move/turn actions

→ **Mainly navigation and viewpoint control, not everyday object interactions**

More reliable and generalizable audio-visual world models

for a broader range of everyday interactions and their sounds

Real first-person recordings
everyday activities

Audio-visual generator
Model how actions and sound relate

Temporal prediction
Add history, memory and action inputs

New environments and tasks
Studying model behaviour for new environment and tasks

Sound in First-Person Video

The camera sees the activity in front; the microphone also hears the wider surroundings



FG = sound of a visible action · BG = ambience or off-screen sound

FG throwing granules into the water → **sharp high-pitched splash**

BG female speech: “una rama de naranja, vale”
male speech: “¿Ahora?”
continuous water splashing and lapping

Joint caption (excerpt)

*Underneath the scene, **[BG_SOUND: female speech: “una rama de naranja, vale”]** … persist. …*
***[FG_SOUND: sharp high-pitched splash]** coincides with the display of the container. …*

A New Dataset and Benchmark

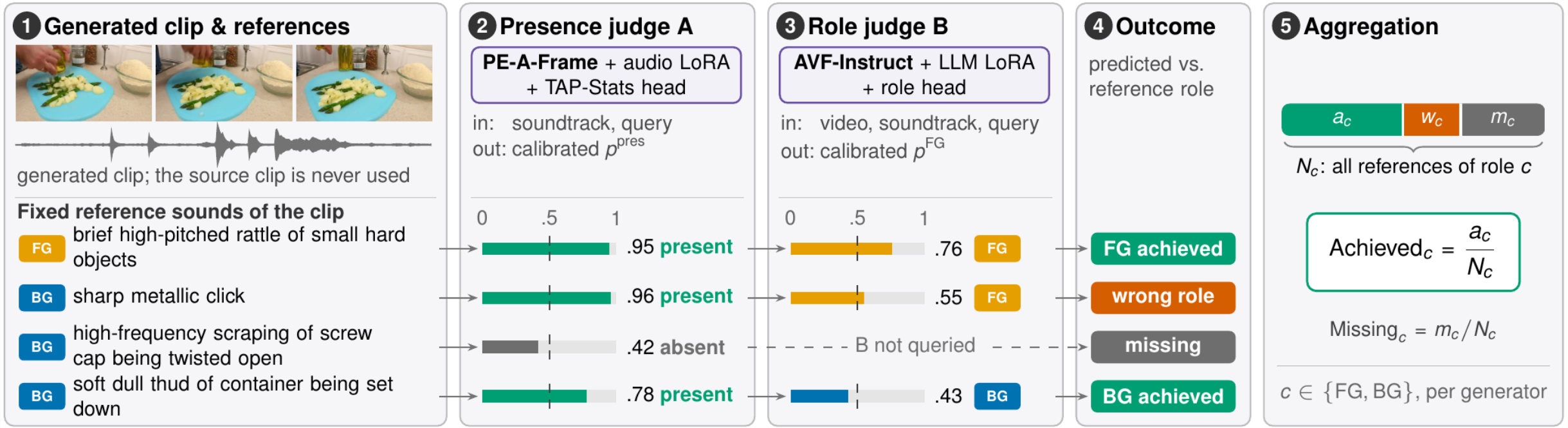
FVD and FAD compare distributions — they cannot tell whether a requested sound appears, or in which role

500 clips
quality + diversity

673 FG + 1,585 BG
fixed sound references

Same prompts
for every model

First frame
for image-to-AV models



Rates are reported separately for FG and BG, each as a share of all references of that role

Conclusions and Future Work

Conclusions:

- We proposed a **MultiAct dataset** (v1, v2) for egocentric long form audio understanding, and two models (**hierarchical SlowFast model** & **hierarchical grammar-tree model**), specifically designed to perceive and process the multi-scale temporal structures inherent in egocentric long-form audio, effectively modeling events, sub-activities, and overarching activities.
- We proposed two datasets (**YT-Speech** & **AV-QTSS-Syn**) for wearable spatial audio generation based on 360 video, and two models (**the localizer-render model**, and **the AV-QTSS model**), with 360° video and the single-source omnidirectional channel as input, or 360° video and multi-source omnidirectional channel as input.
- We also developed a new dataset and benchmark of **world models for understanding the relation between actions and sound**.

Future works:

- We will further improve event boundary accuracy, robustness and scope. Parsing operates on event detector proposals. It can relabel, merge, or suppress segments, but cannot refine their temporal boundaries. The parse inherits errors from imperfect event hypotheses. Current results are shown on MultiAct kitchen procedures; broader settings need validation.
- We will study video-to-overlapping multi-source spatial-audio generation, e.g. generate FOA with overlapping sound sources from 360° video alone; no audio input: sounds, timing and directions all come from the video.
- We will study the relation between actions and sound via egocentric world models.



Thank you for listening

Appendix: AV-QTSS-Syn Resources

Resource	Count	Main sources (datasets)
Audio events	3,904 recordings	FSD50K, EARS, WavCaps, ESC-50, FMA, DCASE 2026 Task 4
Visual objects	3,804 identities	Mainly Open Images; LVIS, Flickr30k
Panoramas	1,984 indoor scenes	360-Indoor
Room acoustics	20,627 FOA RIR entries (26 room variants)	DCASE 2026 Task 4, SurrRoom, TAU-SRIR, Arni
Ambience	105 native-FOA recordings	DCASE 2026 Task 4, EigenScape, TAU-SNoise